

Module 1: Money and Finance

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Module 1 Information

[Lesson 1-0: Overview](#)



Hello everyone and welcome to Module 1 of financial institutions. This module is called Money and Finance. It is going to provide a broad overview of money, the financial system and how this all connects with real economic activity. We will begin by introducing money and the payment system and answer the following questions: What is money? How can we measure it? What is the relationship between the quantity of money and the inflation rate? Next, we will describe the financial system. What is the purpose of the financial system? What is the difference between financial markets and financial institutions? This is a course on financial institutions, so we will want to understand who these financial institutions are and how they add value. In a final step, we will put money and finance together and examine the connection between financial development and real economic activity. Financial development means faster and better money and finance. This can be great for economic growth and risk-sharing, but it can also lead to financial instability and economic crashes. By the end of this module, you should have a big picture understanding of what money and finance is and its potential benefits and costs.

Lesson 1-1: Money and Payments

[Lesson 1-1.1. Money and Its Functions](#)

Learning Objectives

- Definition of money
- Payments system

In this session, we will discuss money and the payment system. This will involve two steps. First, we will define the term money. This seems like a very strange question given that we all use money all of the time. However, what has worked as money has changed a lot over time and is even being debated today. In this session, we will think more carefully about what makes an asset usable as money. In a second step, we will describe the modern payment system which consists of the methods we use to pay for things, and explain how it relates to money. Let's start with my favorite motivating example, the beach vacation. Beach holidays have been a constant throughout human history, well, presumably. How we have paid for these holidays, the money, and payment choices we've made have changed in meaningful ways. Arrowheads in 8th century Japan, seashells in China, salt in Venice and metals such as gold. These are examples of commodity money. Commodity money has been used to pay for things throughout human history.

Commodity Money

- Asset-backed
- Intrinsic value

It has two characteristics. First, it is backed by a physical item. Second, that item has some intrinsic value. The item is worth something even if it is not used for payments. Salt is a valuable preservative. Arrowheads can be used to shoot people. Fast forward about 1000 years to 16th century Sweden. At that time copper plates were used as currency. This was a form of commodity money because copper wasn't all that valuable at the time. This meant you had to lug a large amount of copper around to make transactions, not terribly convenient. In 1657 along comes a Swede named Palmstruch. He sets up the Stockholms Banco which was the first European bank to print paper money. This paper money was redeemable on-demand. You could exchange the paper for the metal money, so your copper, that you had to deposit it at the bank.

Paper Currency (Cash)

- 17th century Sweden

So long as people believed that there was enough copper at Palmstruch's bank then the paper money was valuable. In 1667, however, the bank began to run out of copper. Swedes started to lose faith in Palmstruch and the bank had to close down. The question of why they started to lose faith is absolutely fascinating and central to the topic of bank risk management but that's for a later time. The United States issued paper money backed by gold until the 1930s. This was called the gold standard.

Paper Currency “Cash”

- Government-issued fiat money

Today we use fiat money which is paper money that is not backed by any physical commodity. The paper money has zero intrinsic value. In 2021 it costs the US Treasury just 14 cents to print

and engrave a \$100 bill. Why is fiat money useful for anything? Well, it's because the government says so. The law says private businesses must accept dollar bills as payment. I'm also allowed to pay my taxes in dollars so long as I have trust in the government then these pieces of paper are valuable.

Plastic Cards

- Debit and credit cards

We all know what a debit or credit card looks like. You probably have one within arm's length. But you might not understand the mechanics of how this plastic works. A debit card payment provides instructions for my bank to send my money from my bank account to a merchant. Without a bank account containing money, this just wouldn't work. On the other hand, when I use my credit card I'm actually getting a loan from a bank, when I swipe my credit card I am authorizing my bank to send some of its funds to pay for my beach vacation, and then I can repay my bank at some later date. The first modern plastic credit card was introduced by the Bank of America in Fresno California in 1958.

Plastic Cards

- Debit and credit cards
- Online payments

Nowadays smartphone applications and other online payment systems are replacing our need for plastic debit and credit cards. Apple Pay, Venmo, WeChat, these are 21st century systems that allow us to make cheap, quick, and secure money transfers to friends and family or payments for our shopping. We've come a long way from sea shells to PayPal.

Money

- “Generally accepted means of payment”

Note: money versus wealth

Now let's take a step back and formally define what money is. It is a specialized asset. It is generally accepted as a means of payment for goods and services or the repayment of debts and this includes paying taxes. The most important examples of money are the cash in your pocket and your bank account balances. This is what I want you to keep in mind. I also want

you to keep in mind that money is not the same as wealth, a wealthy individual might have lots of assets besides money. In fact, if you think of your favorite rich person they probably have lots of houses, yachts, rocket ships, stock options, and so on. None of that stuff is money.

Money as a Means of Payment

- Finalizes transactions
- Value is understood

Money is a means of payment. This is money's special role. In modern economies transactions are frequent. They can be spread out across long distances. Money is central to a well-functioning economy. We use money to finalize transactions. I don't have to wait for payment later. I get my money in hand, exchange complete. The value of money is well understood. I don't require any information about the value of money from you. I don't need to check the newspaper or online for recent transaction prices. I know exactly what is worth once you have paid me.

Two Natural Implications

1. Money is a unit of account
2. Money is a stable store of value

If money is a means of payment then two outcomes follow quite naturally. First, we say that money is a unit of account. This means that we can quote prices and record debts in terms of it, in terms of dollars and cents. This allows us to easily compare the value of different products. Second, money must have stable values. If we are going to use money in lots and lots of transactions then it must hold its value from day to day. Its purchasing power must remain relatively stable over time. If its value were unstable you may have heard of hyperinflation where the purchasing power of a currency rapidly declines over time then merchants might not be willing to accept money. This is why in some countries we can use the local currency as well as US dollar as money. Merchants understand that the US dollars are a stable store value.

Lesson 1-1.2: The Payments System

Payments System

- “Financial arrangements that facilitate transactions”
- Cash versus noncash payments

Money in the payment system go hand in hand. Formally, the payment system is a web of arrangements that allows for the exchange of goods, services, and assets. It's often called the financial plumbing of the economy. These systems require more infrastructure than we perhaps realize and are a source of rapid behind-the-scenes financial innovation. The economy depends on an efficient payment system so governments and policymakers pay very close attention to it. In many cases, central banks such as the Federal Reserve System were created to facilitate payments. It is important to recognize that not all transactions involve an exchange of paper currency. Sometimes we make non-cash payments. All non-cash payments eventually involve an exchange of money. Usually this will be from one bank account to another bank account but there can be several steps before that actually happens.

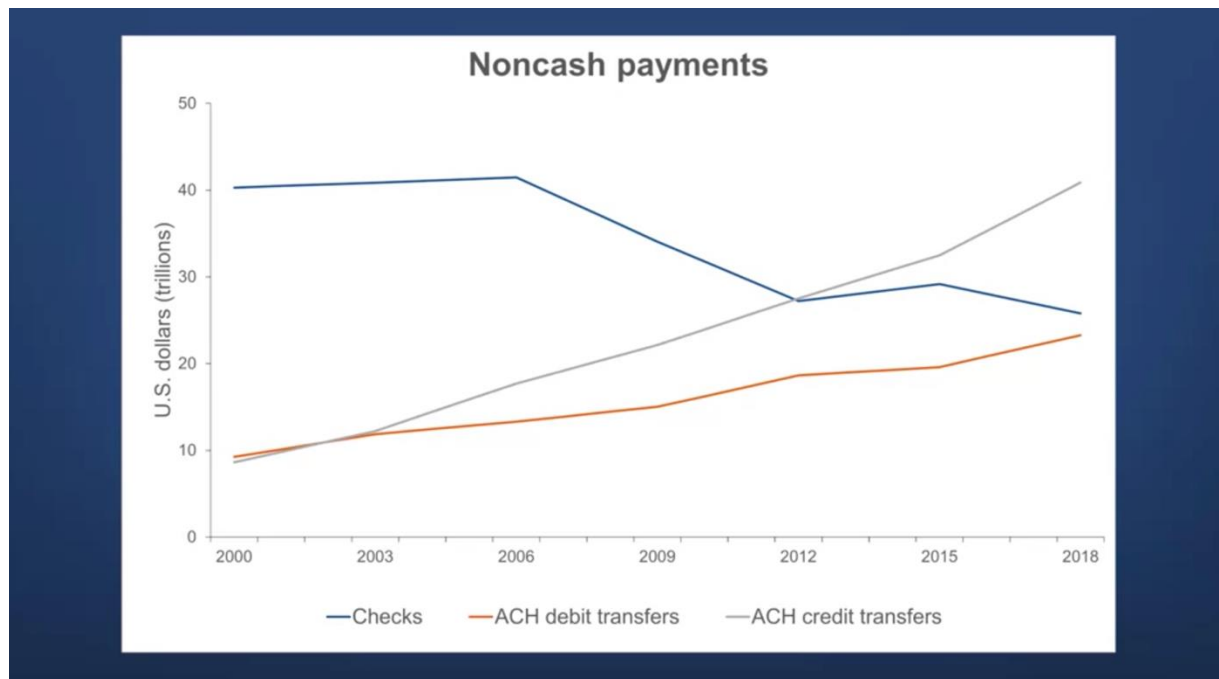
Noncash Payments

- Paper checks
- Automated clearing house (ACH) transfers
- Debit and credit cards
- Digital wallets and apps

Here are the main types of non-cash payments. Believe it or not, paper checks are still an important payment method in the United States. I will say that again in case you missed it. A paper check is not money. It is not currency. It is a method of payment. When you give someone a check to pay for something, this check provides an instruction to your bank to move your funds from your bank accounts to a merchant's bank accounts. In the U.S., the local Federal Reserve is often responsible for clearing these checks, which means processing the transfer of money between bank accounts. The other common payment types are all electronic in nature. Automated Clearing House or ACH transactions act just like checks. An instruction is given from your bank to move money from your bank account into another bank account but they happen electronically and are faster and have less administrative hassle. ACH transfers are most often initiated with a debit or a credit card. With a debit card transaction, funds are transferred from my bank accounts. With a credit card transaction funds are transferred from the credit card issuing banks' accounts. A loan is created in the process. Prepaid debit cards are important means of payment. In some countries, they are not linked to a customer's bank account. Digital wallets allow us to initiate electronic debit or credit transfers from our computers or smartphones. They most often store our payment information from our debit or credit cards. Apple Pay is a market leader in this space. Other money transfer apps such as Venmo or Zelle allow money transfers between friends and family.

To summarize, we can make payments using cash or a non-cash payment method. These non-cash payment methods are often linked to the funds in our bank accounts. This is why economists consider paper currency plus bank account balances as a core measure of money. According to data from the Federal Reserve, in 2018 there were

about 175 billion non-cash payments with a total value of about \$100 trillion. Before you ask we do not have any information about cash payments.



Here we can see the breakdown in the value of non-cash payments by the type of method used, so this is not the number of payments but the dollar value of payments. We can see the checks and then the electronic payments. Clearly, the trend is away from checks and towards electronic payments. In 2012, credit card payments exceeded checks for the first time. Although I must say, checks continue to play a more important role in payments than I would have thought. This is a case study that might make some of you shout at me – Is Bitcoin money?

Is Bitcoin Money?

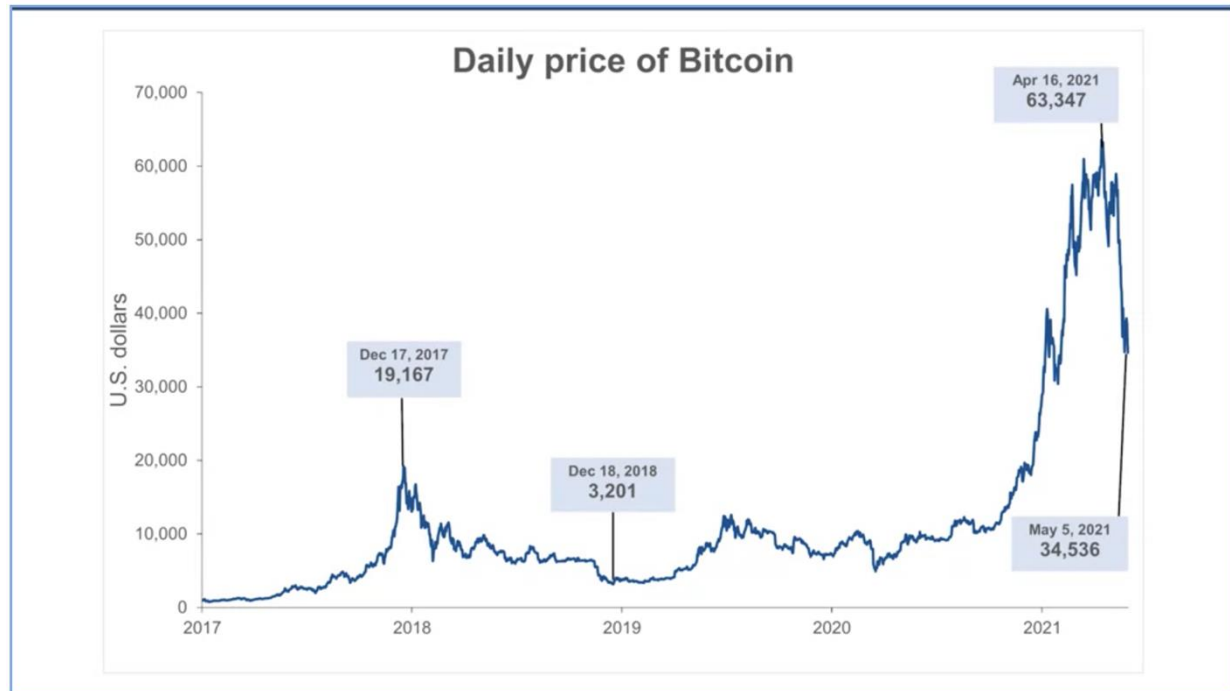
- Crypto = private digital currency + blockchain
- Potential advantages

Cryptocurrencies combined new payment systems with new currencies that are not issued by governments. The currency itself is a digital token that can be bought and sold. Trades are recorded using distributed ledger technology also known as blockchain. This involves an electronic database that records and verifies transactions without the need for verification by a government authority. Bitcoin, which has been around since 2008, is by far the most famous example. Now there are two potential advantages here. First, the value of cryptocurrencies cannot be undermined by governments since they do not control the supply of these digital tokens. Second, since transactions do not need to be verified by any government authority, users can make electronic payments while staying anonymous, so is Bitcoin money or not? Well, let's analyze our three characteristics of money.

Is Bitcoin Money?

1. Means of payment? No
2. Stable store of value? No
3. Unit of account? No

Is Bitcoin a useful means of payment? Well not really. It has no intrinsic value since it is not commodity money backed by an asset. It is not legal tender either, so businesses are not required to accept it. It cannot be used to pay taxes. Now, this might change but we're not quite there yet. So far, Bitcoin plays no role in any normal economic activity in any well-functioning country. If anything Bitcoin transactions have been associated primarily with criminal activity such as drug dealing or cyberattacks. For this reason, what most societies don't want to make life easier for criminals. China has clamped down on crypto transactions. Next, is Bitcoin a stable store of value? No, it doesn't really look like it. These big swings in prices make Bitcoin resemble a speculative asset rather than a currency. For this reason under the US tax code, Bitcoin and other cryptocurrencies have been classified as property and taxed as an investments.



Given the wild fluctuations in its value, the prices of taxi rides or Tesla cars would be highly unstable if they were priced in terms of Bitcoin as opposed to US dollars. Perhaps this might change in the future, but for now, Bitcoin does not resemble a reliable government-issued currency such as the U.S. dollar or the Euro.

Summary

- Money as a means of payment
- Payment system facilitates transactions

Let's sum up. Money is a special asset that is widely accepted for making payments. It's useful for measuring and comparing values, and it's a store of value that allows me to

transfer my purchasing power into the future. The payment system sends our money around. We make the distinction between cash and non-cash payments. Recent data seems to indicate that electronic payments are becoming more important.

Lesson 1-1.3: Money and Inflation

Learning Objectives

- Measuring money
- Money growth and inflation

In this session, we will move beyond money as a concept and towards the measurement of the total amounts of money circulating in the economy. Why do we need to measure money? Well, there is evidence that too much or too little money can affect economic growth, interest rates, and inflation. As a result, central banks and policymakers have paid close attention to the money supply as well as innovations in money and the payment system. In this session we will examine the relation between aggregate money growth and the inflation rate in the past 50 or so years.

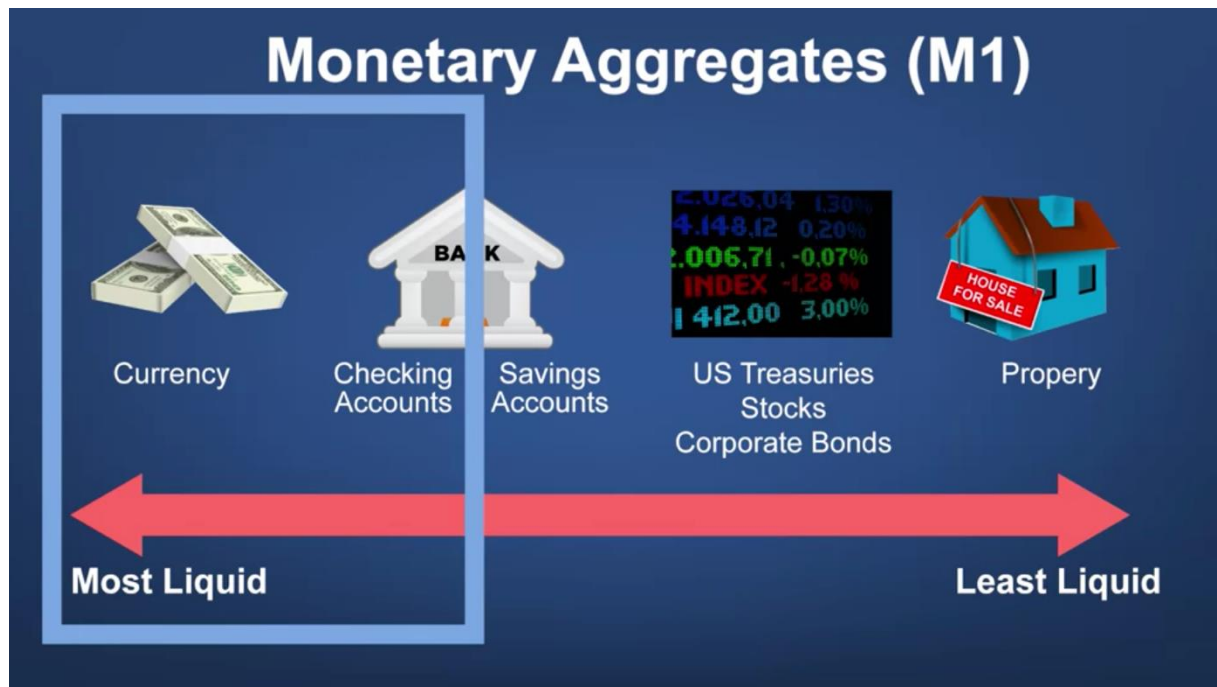
Money and Liquidity

- Money is used for payments
- Liquidate asset into money

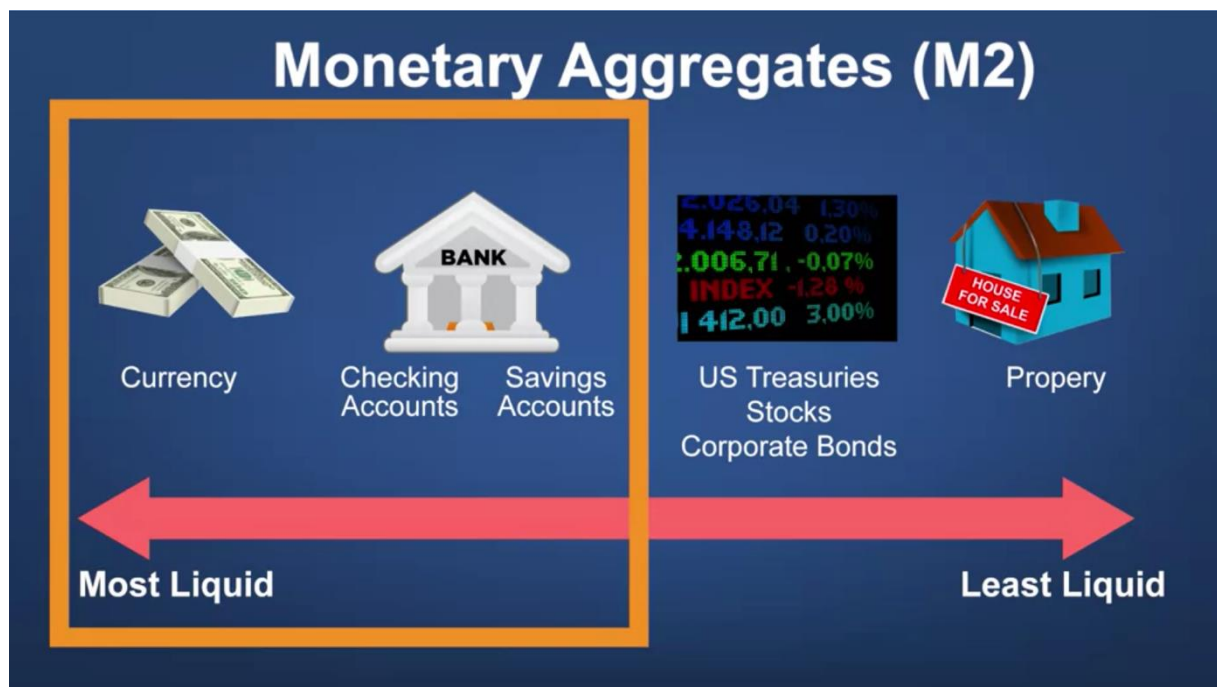
Money is a special asset that we use for payments. It is a subset of our wealth. We use it for transactions to make purchases or pay down our debts. Money is the most convenient way to pay for things. But of course we hold our wealth in a variety of other forms. We have funds in bank accounts, we invest in stocks and bonds, and we may own a house. We can use these other assets for payments, but these other methods can be more expensive. Why? Well, we first have to convert the asset into money. This is called liquidating the asset. Asset liquidity tells us how costly it is to convert an asset into money. Cash is already money, so it is the most liquid asset. Some bank accounts allow us to write checks or charge debit cards. These accounts work just like cash as we all know. We can rank all assets by their degree of liquidity.



Cash is the most liquid asset since it can be used right away as a means of payment. Houses and other property such as jewelry or family heirlooms are valuable, but they might take time to sell at the right price, so they are considered the least liquid category of assets. Securities and savings accounts with lockup periods sit somewhere in the middle. We can use this asset liquidity ranking to arrive at a definition of money. There really is no ideal way to do it and the definition has changed over time, including only cash as money is probably too strict and including stocks and bonds is too loose since people probably don't use stocks to pay for groceries.

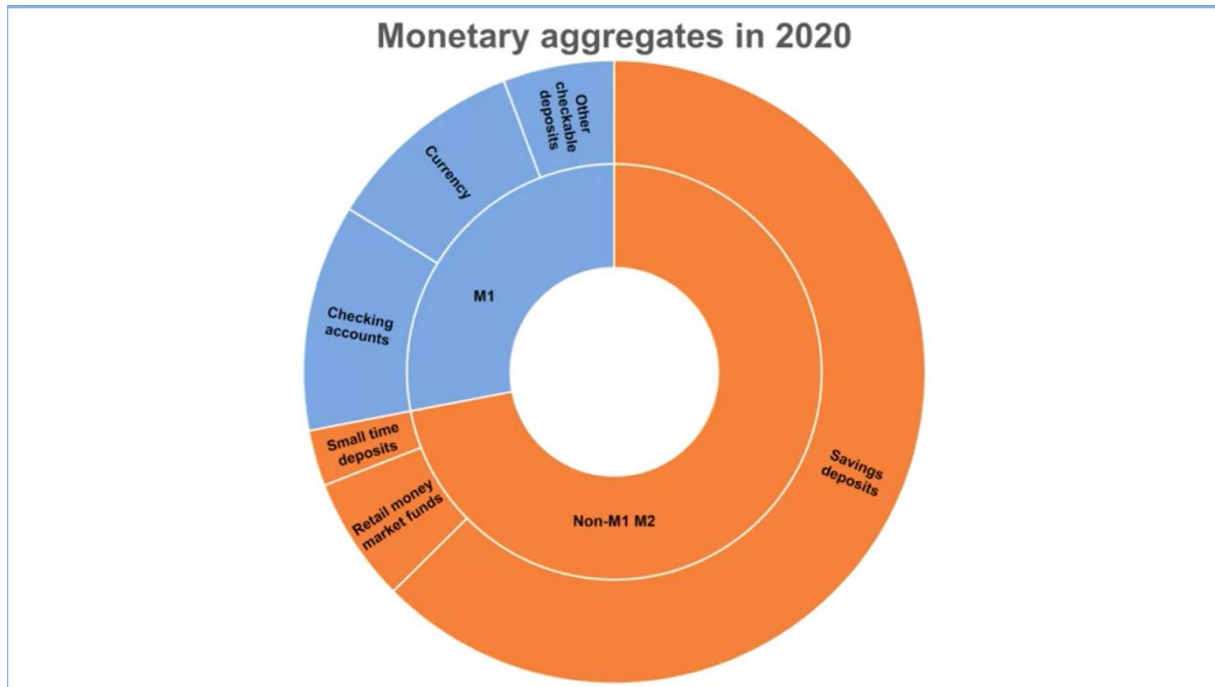


In practice, economists work with two basic definitions of money, M1 and M2. M1 is the strictest definition. It includes only paper money that is in the hands of the public. That's in our wallets not sitting in bank vaults, as well as checking accounts at banks.



M2 is a broader measure of money. It takes that M1 and adds in some savings accounts. The savings accounts can be located at banks or a type of mutual fund called a money market fund. These are highly liquid accounts, but they are not quite as convenient as checking accounts. For example, bank savings accounts may restrict

customers to a certain number of withdrawals per month. Since savings accounts pay interest, households have moved their funds from M1 into M2. This means that M2 moves more closely with economic activity, and so economists tend to focus on this measure.



Here is some data from the Federal Reserve on monetary aggregates in April 2020. We have a breakdown into M1 and then M2. M1 includes currency plus checking accounts. M2 is the whole pie, so it adds in the savings accounts. As you can see, if we just focused on currency as our measure of money, then we miss a big part of the picture. That's why economists prefer to look at M2.

Inflation

- Increasing prices
- Measurement: consumer price index

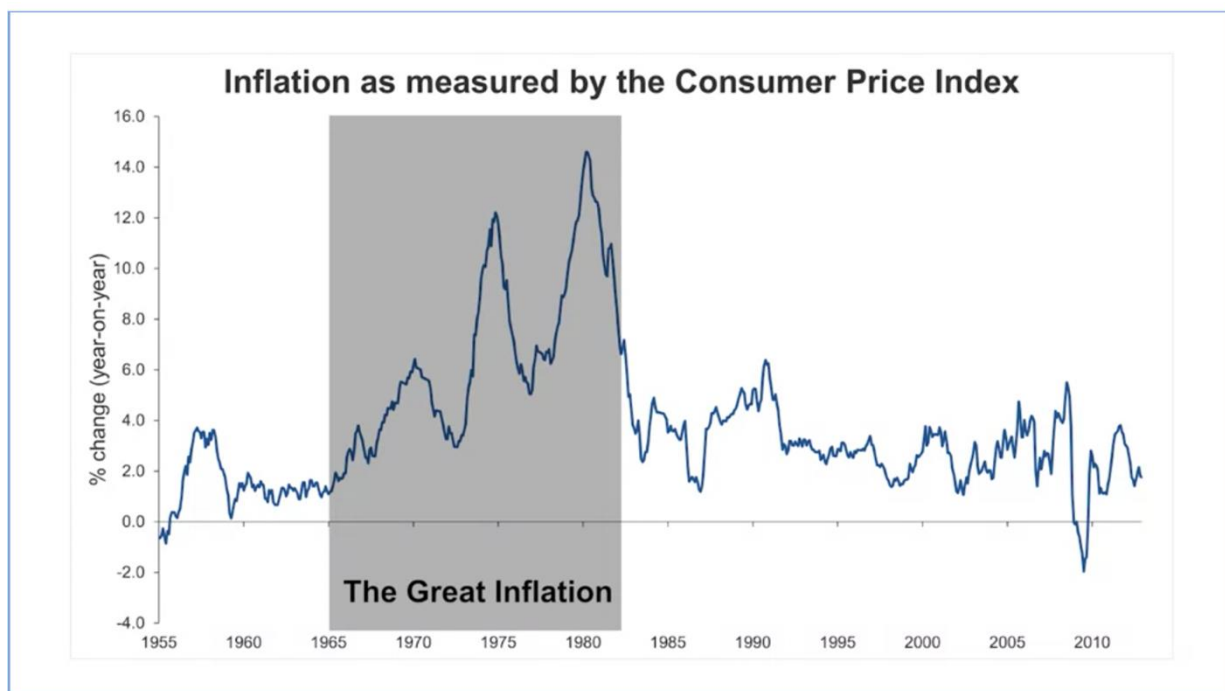
Let's now introduce the concept of inflation. Inflation captures the rate at which prices increase over time. With high inflation, we must spend more dollar bills to buy the same basket of goods as last month. Put differently, with high inflation, money becomes less valuable. The consumer price index or CPI is the most commonly used measure of the inflation rate. It measures changes in the price of the same basket of goods and services: food, housing, cell phones, and so on, over some fixed period of time. In January 2021, the 12 months CPI in the United States was about 1.4 percent. This means that I would pay \$101.40 to buy the same basket of goods that cost me exactly \$100 to buy one year prior in January 2020. Growth in the CPI tells us about the purchasing power of our money. In the past, there have been several episodes of very high inflation. Germany in 1923, the daily inflation rate was 21 percent. This meant that prices doubled every three days. We saw similar outcomes in Hungary in 1946, Zimbabwe in 2008, and Venezuela in 2018. Why does this happen? The usual observation is that the government finances its deficits by increasing the money supply, by forcing the central bank to print money to pay for its spending.

Money Growth and Inflation

- Hyperinflation
- Money growth proximate cause
- Economic collapse

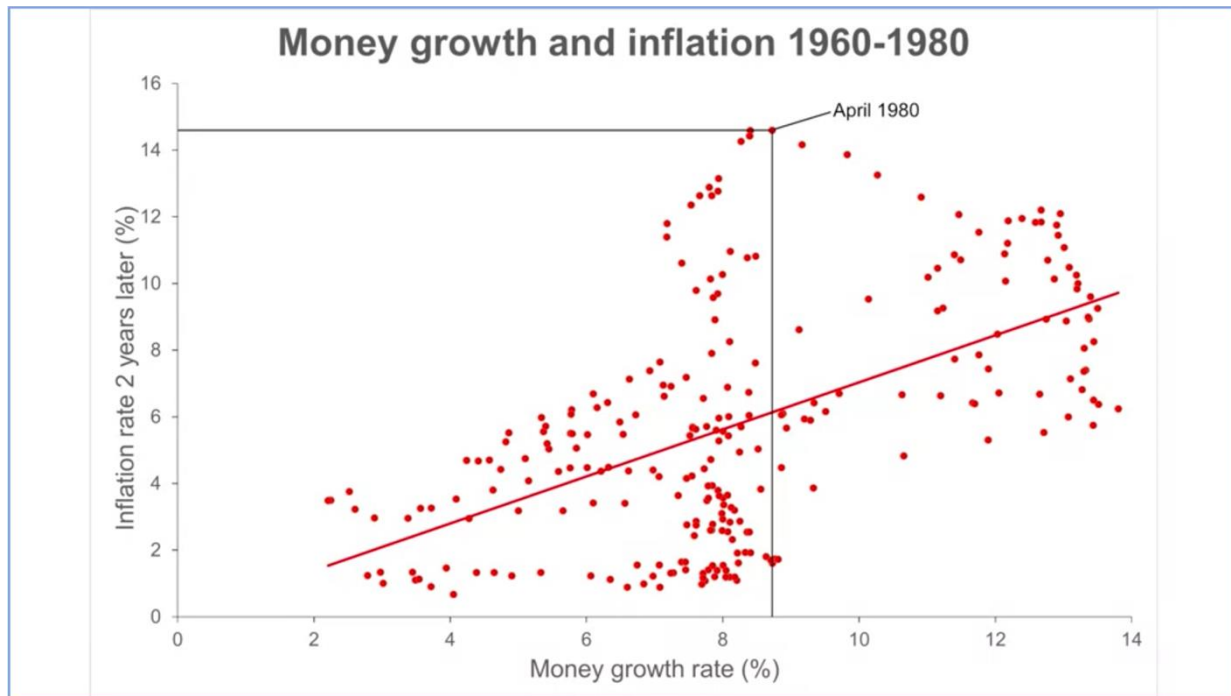


These periods of very high money growth are then followed by periods of very high inflation. When there is more and more money chasing a fixed number of goods, money becomes less valuable, terrible, dire outcomes for households and businesses usually follow. The experience in the United States in the 20th century at least has never been that bad.

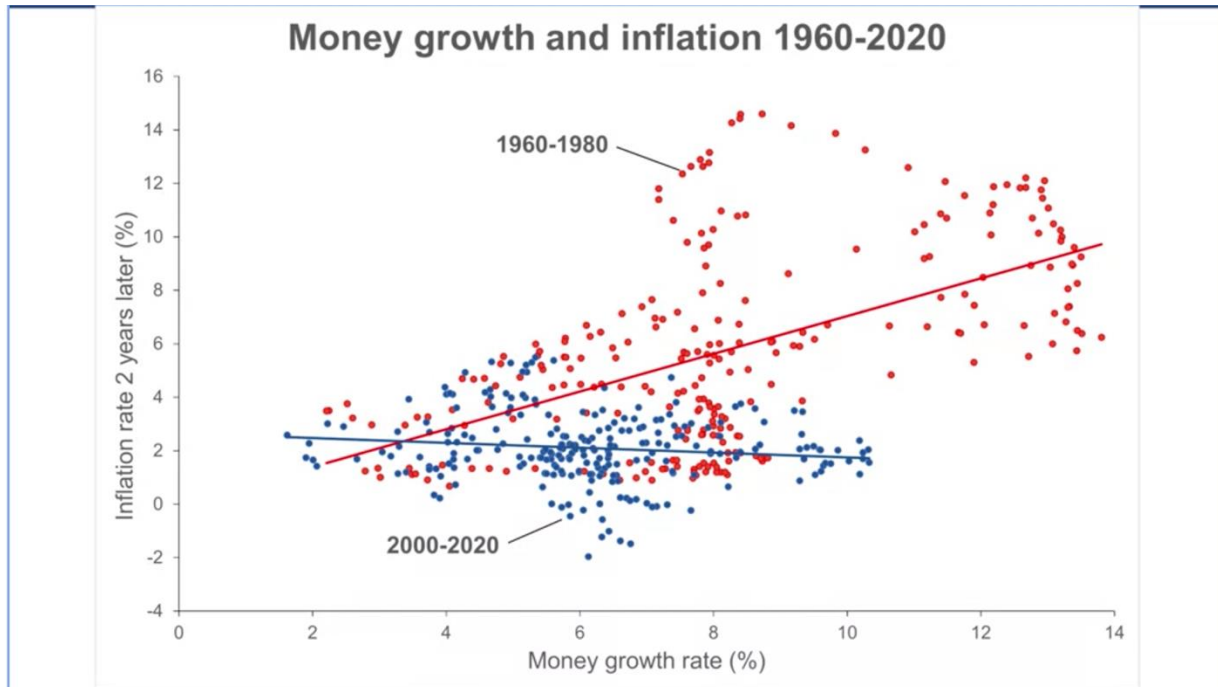


Here is a plot showing monthly CPI growth from 1950 to present. Inflation has generally

been below five percent except in the 1970s and 80s when inflation exceeded 10 percent. This period has been called The Great Inflation.



At this point, we're going to take two steps. First, let's use our monetary aggregates to examine whether money supply grew before the Great Inflation. Second, let's see if our monetary aggregates can forecast inflation in the 21st century. Here is the relation between inflation and money growth from 1960 until 1980. Inflation is measured using the CPI, money growth is measured using M2. Money growth is lagged two years to make sure we get the timing correct. Does money growth in 1960 have an association with inflation in 1962? And so on. In the graph, each red dot represents a month. This red dot here represents April 1980. At this point in time the inflation rate was around 15 percent and the money growth rate from two years prior was around eight percent. The fitted line shows a positive association between money growth and inflation. Periods of high money growth are associated with high inflation two years later. Periods of low money growth are associated with low inflation also two years later. This fits our intuition. But the story doesn't end there.



If we, instead, focus on the post 2000 period, we lose the correlation between growth rates in M2 and the CPI. M2 is no longer useful for predicting inflation in the 21st century. Why is this the case? There are two potential explanations. First, the relation between money growth and inflation does not show up in a low inflation environment. You may have noticed that after the year 2000, the inflation rate is hovering around two percent. That is abnormally low by historical standards. The second possibility is that M2 is no longer an accurate measure of money. In the 21st century, there are much lower transaction costs in financial markets, credit cards are widespread, and new digital payment technologies are popping up every day. The way we use money and make payments has changed quite a bit, and so perhaps we need a new measure of money.

Summary

- Monetary aggregates: M1 versus M2
- Money growth and inflation (?)

Let's wrap up. In this session, we went from the concept of money to its measurement at the economy wide scale. We used these monetary aggregates to try to understand the relation between money growth and inflation. During The Great Inflation of the 1970s and 80s, we saw that money growth was also elevated. Finally, in the 21st century, we saw this relationship became murky. This raises questions about how we should measure money in the future. It also cast doubts on whether past approaches by governments and central banks to control inflation will be effective in the future.

Lesson 1-2: The Financial System

[Lesson 1-2.1: What Does the Financial System Do?](#)

Learning Objectives

- Purpose of financial system
- Common financial instruments

In this session, we will discuss at a very high level what the financial system does. We will begin by introducing borrowers and savers. We will then examine how funds flow through the financial system and how these funds are exchanged for financial instruments.

Meeting Financial Needs

- "Borrowers"
- Internal funds
- External funding from "savers"

Our starting point is an economic agent that has a funding need. This could be a business with a profitable investment opportunity. Now, this business could dip into its cash reserves or even sell assets to fund this new investment. However, these internal funds might not be enough. In this case, it might make sense for the business to be a borrower and raise external funds from savers.

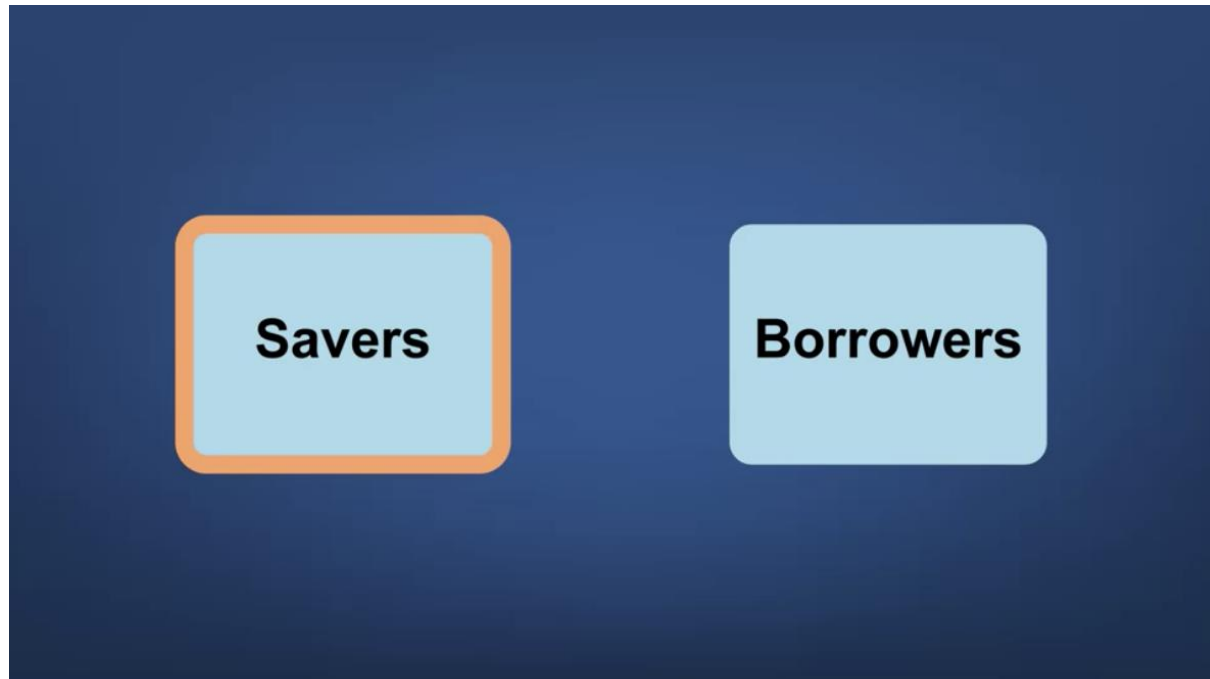


Savers

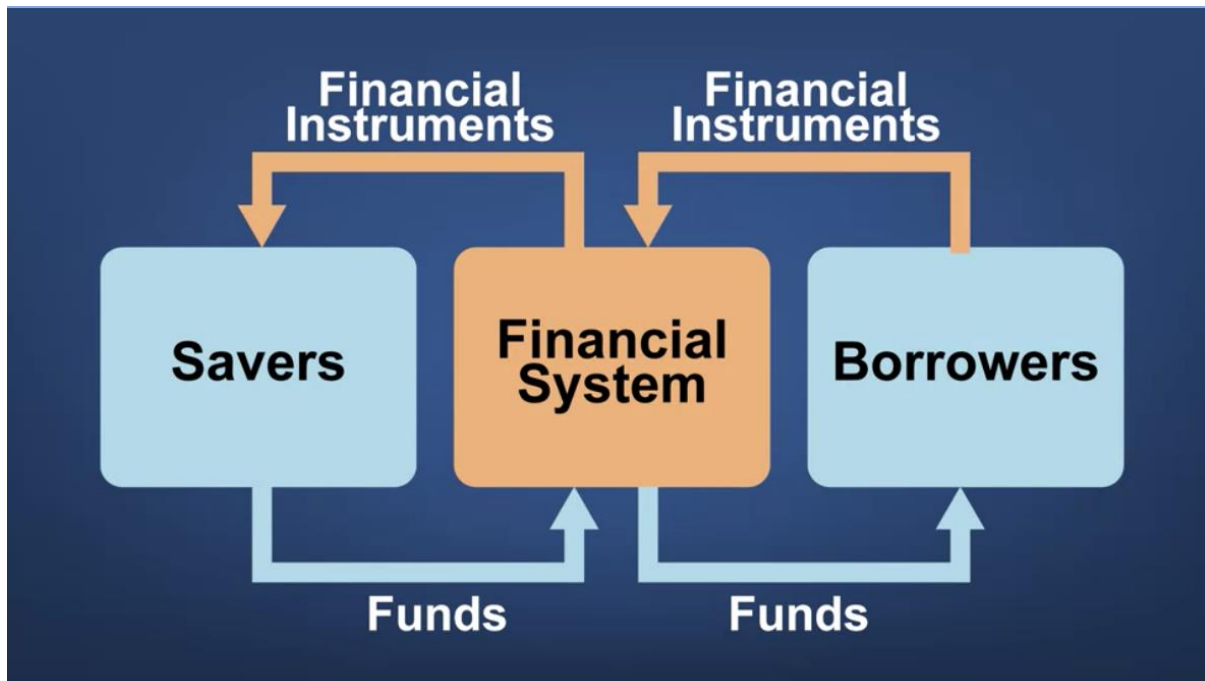
Borrowers

Here are our savers and our borrowers. Borrowers can be just about anyone. Households that need funds to buy a house or something at the shops. The business

that I just mentioned may have a short-term working capital need or require longer-term funding to expand operations. Farmers might need money to buy tractors. And, of course, the government might need to borrow to fund public goods.



Savers are mostly households with more income than current spending needs. The savings can be put towards retirement. This often happens through a pension fund or a taxable mutual fund. They can be put in a bank, which is also useful for managing cash. Rather than spending more today, households can save for a rainy day by buying insurance.



Now here comes the financial system. At a very high level, and this might sound weird, the financial system connects savers and borrowers. By connecting them, the financial system channels savings to meet the funding need of the borrowers. The end game here is economic prosperity. The financial system channels resources to their most efficient uses, and this leads to economic growth. In exchange for their funds, savers receive these financial instruments. Let's talk about them in more detail.

Financial Instruments

- Written legal obligations
- Transfer of funds
- Conditions

A financial instrument is basically a formal commitment to repay. I give you some money. You have to repay me next week. We can be more precise. It is a legally binding obligation that courts and governments will enforce if something goes wrong. It requires that funds are transferred between participants under certain circumstances. I am repaid fixed amounts on certain dates. I am paid a lump sum if I accidentally crash my car and so on. That's quite abstract so let's consider some examples that we all know.

Savers

- Bank deposit
- Mutual fund share
- Insurance contract

Let's start with the savers. We are all familiar with a bank deposit account. As a saver, we put our cash in the bank. These funds are redeemable on demand. This means we can get our cashback whenever we want. Bank deposits are also safe and convenient. Instead of paper money, we can use checks, or debit cards, or mobile banking to make payments. With a mutual fund share, I exchange my money for a share in an investment portfolio. This helps me store and maybe even grow my wealth into the future. Insurance is another form of saving. Let's say I'm worried about a car accident. What can I do? I give up some spending today and pay an insurance premium, and then in the bad state of the world when I crash my car, I get a payout. This insurance payout covers the cost of fixing my car. Insurance is a type of financial instrument that allows our savers to transfer risk.

Borrowers

- Equity

What types of financial instruments do borrowers use? Equity. A business owner can sell shares. This will pull in funds today. The buyer of the equity now has a stake in the firm. This entitles the buyer to a share in the future profits of that firm.

Common Financial Instruments

- Equity
- Bond, bank loan, mortgage

When we think of borrowing, we usually think of debt, bonds, bank loans, mortgages. Here a borrower usually gets a lump sum today and then promises to repay in the future. The payments are usually fixed amounts that happen on fixed dates. That's why these financial instruments are sometimes called fixed income. Government bonds used

to have coupons attached to them that could be torn off and redeemed at a local bank for a cash payment. There can be strings attached. With a residential mortgage, that's when you get a loan to buy a house. The borrower usually posts the land and the house as collateral.

Common Financial Instruments

- *Savers*
 - Bank deposit
 - Mutual fund share
 - Insurance contract
- *Borrowers*
 - Equity
 - Bond, bank loan, mortgage

These are all financial instruments. They are all invention, sometimes called innovations of the financial system.

Financial Instruments Meet Specific Needs

- *Savers*
 - Convenience
 - Store wealth
 - Risk transfer
- *Borrowers*
 - Long-term funding

You might be thinking, why do we have so many financial instruments? Well, they are tailored to meet the demands of savers and borrowers. As savers, we value convenient ways to store and grow our wealth. Just hanging onto money probably doesn't make sense for most of us. We may also be afraid of some adverse events and want to buy insurance. Borrowers want long-term funding. They might be happy to raise money by issuing debt. Some business owners might prefer to cash out by selling equity instead.

Summary

- Financial system
- Financial instruments

Let's sum up. The financial system sits between our savers who have funds to spare and our borrowers who have an investment need. The financial system channels those

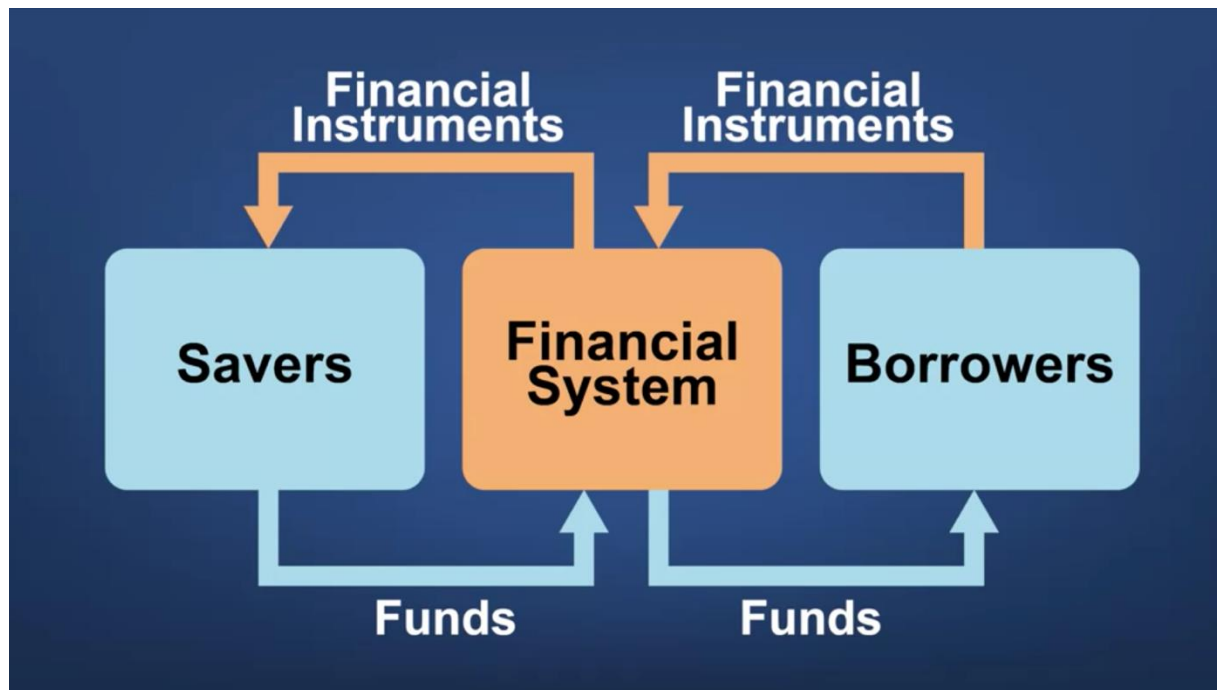
funds from the savers to the borrowers. It exchanges funds for financial instruments. These financial instruments are designed to meet the needs of savers and borrowers.

Lesson 1-2.2: Financial Markets Versus Financial Institutions

Learning Objectives

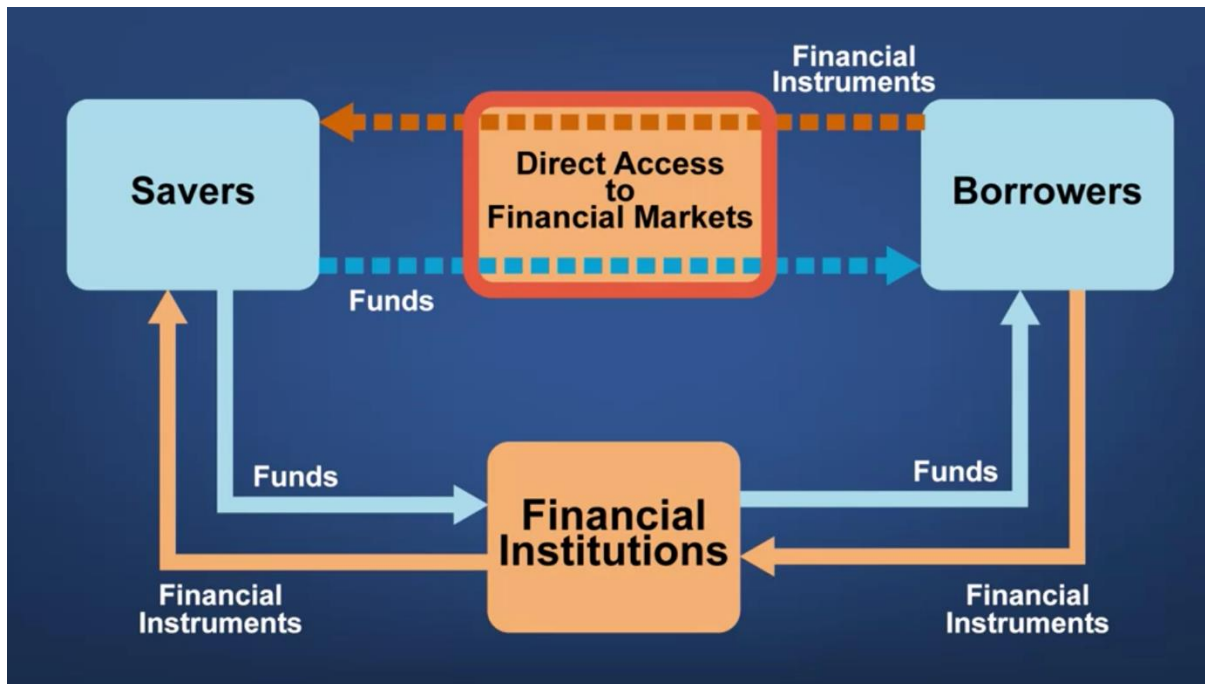
- Financial markets
- Financial institutions (e.g., banks)

In this session, we will discuss how funds flow through the financial system. We will introduce and characterize financial markets, we will then explain what financial institutions are and how they add value. To keep things simple, we will focus on a bank.



The financial system channels funds from savers to borrowers in exchange for financial instruments. But who or what is this financial system? That's what we are going to work on right now. It turns out that there are two ways that funds flow through the financial

system, either directly through access to financial markets or indirectly via financial institutions. In both cases, borrowers receive their funds and savers receive their financial instruments, but what's interesting is just how that exchange takes place and who is involved.



Let's start with financial markets. How can a saver participate directly in financial markets? Well, let's say you wanted to buy a share of Apple. Typically you will not buy the share from the company itself. Instead you would buy the share from another investor on the New York Stock Exchange.

Direct Exchange via Financial Markets

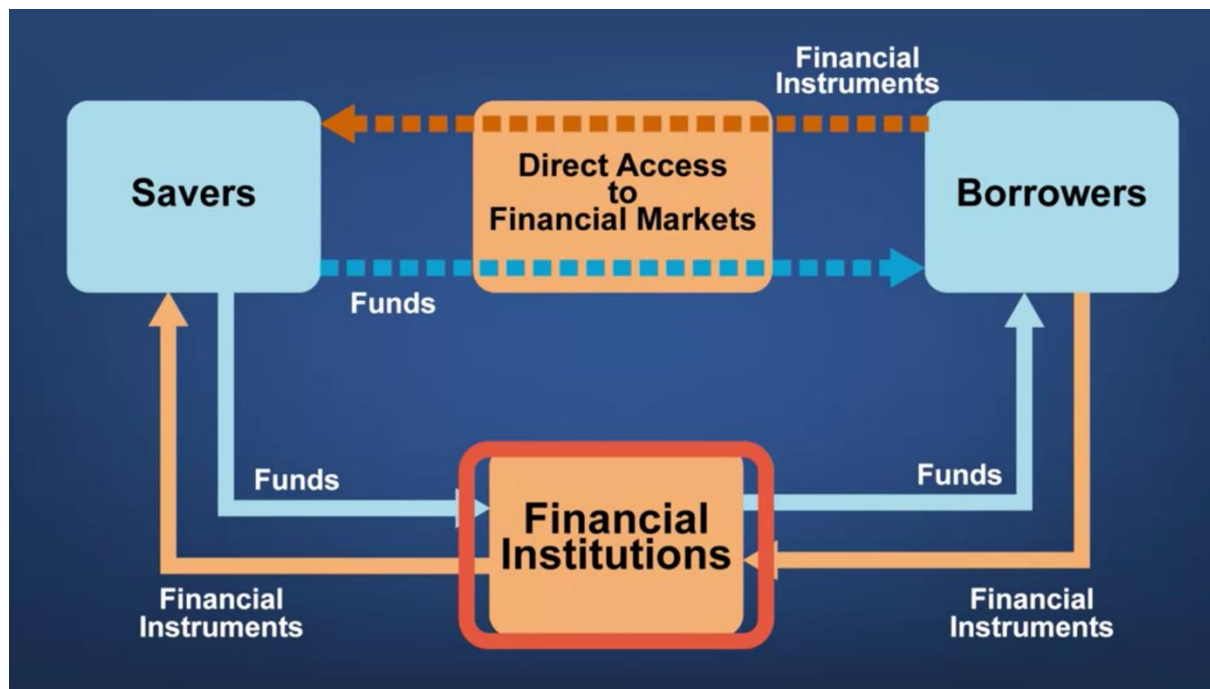
- NYSE: AAPL
- Brokers and dealers

That's why we denote Apple with this ticker symbol. This type of transaction is what we call direct exchange via financial markets. On the direct exchange parts, while we say direct exchange, this type of transaction will usually be executed with the help of a financial firm called a broker or a dealer. For a fee, a broker will find your trading partner, on the other hand, but still for a fee a dealer will act as your trading partner selling you a financial instrument that they carry in their inventory. Nowadays most of this interaction happens electronically. For example, you may have heard of E*Trade. This is an online trading platform that offers both broker and dealer services for retail investors like you or me. There are many types of financial markets. We classify them in terms of the type of financial instrument that can be bought or sold. For example, stock or equity markets are where the shares of businesses are traded. Debt markets involve loans, mortgages or corporate bonds. Money markets are where short-term bonds. Those with maturity less than one year can be bought or sold. To make this concrete, when Apple wants to fund its operations, it can borrow by selling equity long-term bonds or short-term bonds known as commercial paper. After issuance, the equity will be traded in a stock market, the long-term bonds will be traded in a bond market and the commercial paper will be traded in a money market. We all know that well-functioning financial markets play a central role in modern economies.

Modern Financial Markets Have:

- Low transaction costs
- Accurate prices
- Investor protection

Countries with the largest and most active markets all show three qualities. First, low transaction costs. It is cheap for borrowers to issue securities and it is also cheap for investors to trade among themselves. Second, market participants should be allowed to collect, process and communicate information about borrowers. This will ensure that market prices accurately reflect potential risks and returns. When prices are accurate, resources will flow to their best use. Third, investor protection. Here I don't mean that investors are protected against losses if an investment doesn't work out. No. I mean that savers need to be confident that borrowers will not rip them off. Financial instruments are legally binding commitments to repay and modern financial markets require regulators and courts to enforce contracts. When investors are protected against abuses, they will be happy to invest.



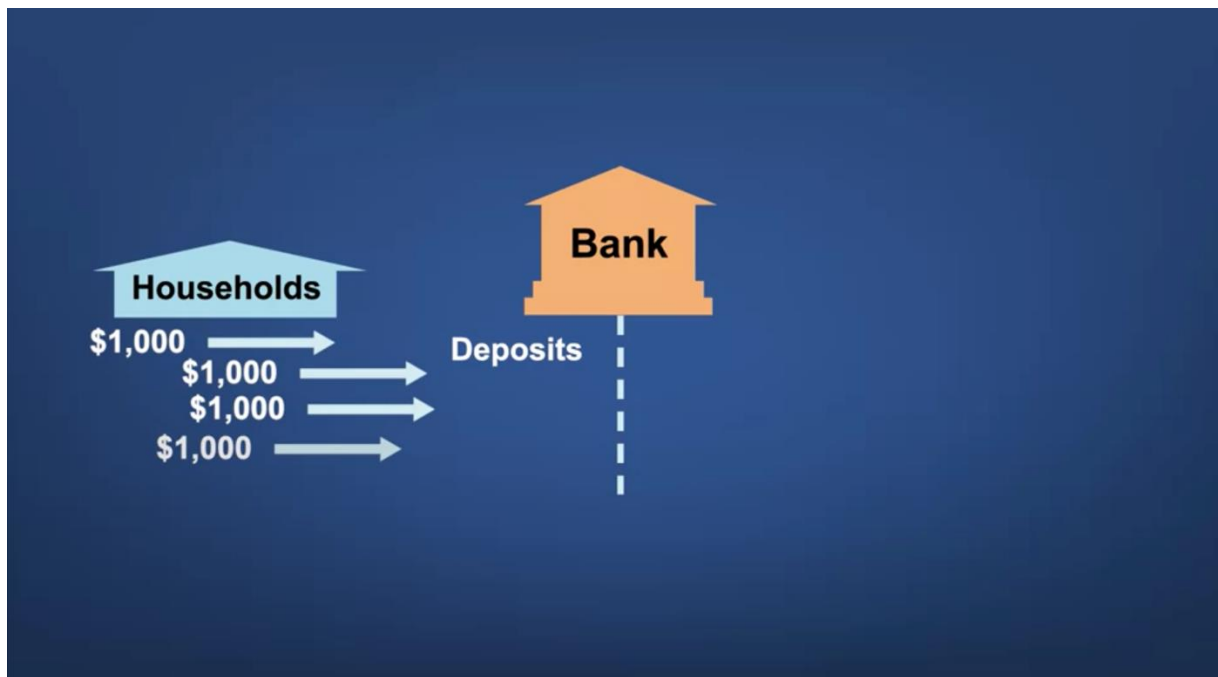
Next, let's introduce financial institutions. Our goal here is to understand what financial institutions are and how they add value over and above financial markets.

Financial Institutions

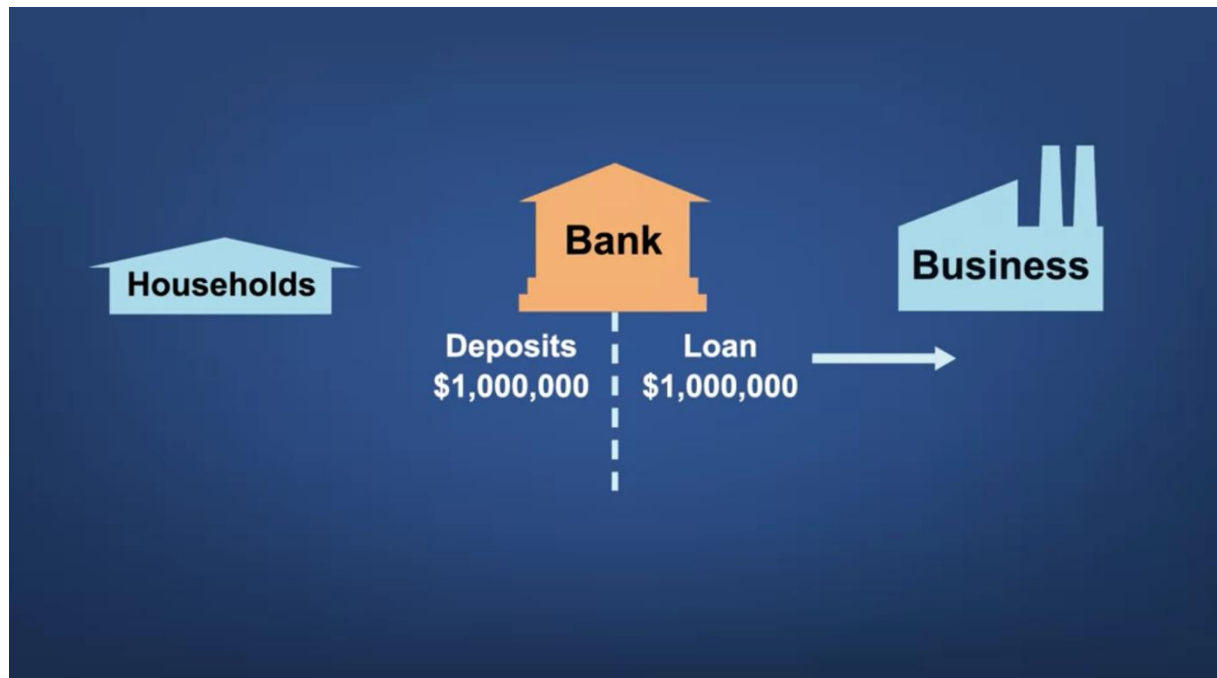
- Firms that channel funds from savers to borrowers
- Classic example: a bank

Financial institutions are firms that channel funds from savers to borrowers. How do they do this? Well, they first raise funds from the savers. They take those savings and usually provide a short-term liability in exchange. They then take those funds that they've raised from the savers and they send them to the borrowers. By this, I mean the financial institution will invest in financial instruments issued by the borrower. At the end

of the day, we still have funds being sent from savers to borrowers but as you can see there is this extra step in the process. This is quite abstract and I'm going to make it concrete in a second, but one final point, a key difference between these financial institutions and the brokers and the dealers mentioned before is that these financial institutions have balance sheets and often make long-term investments. The brokers merely connect savers and borrowers.



Let's work through the classic example of a financial institution. If you can follow this, then you are in great shape. How does this process work at its simplified bank? I am the saver, I put \$1,000 of my savings in a bank in exchange for a deposit contract. The deposit contract is a liability of the bank. It owes me \$1,000 on-demand or whenever I want it back. The bank does this many times over, it pulls one million dollars in savings.



Next, the bank identifies a credit worthy business and lends that business one million dollars as a long-term business loan. That business loan is an asset of the bank. The business owes the bank one million dollars. Putting this together, our bank has channeled funds from savers to borrowers. Perhaps less obvious, the bank has created a balance sheet along the way, a one million dollar loan funded by one million dollars of deposits.

Value-added of Financial Institutions

1. Maturity mismatch
2. Price risk
3. Information

Why do we need financial institutions? What is the point? We know that financial markets and say banks co-exist, so there must be good reasons to have both. Economists have identified at least three problems that financial institutions help to solve. First, maturity mismatch, second, price risk, and third, information asymmetry.

1. Maturity Mismatch

- Borrowers prefer long-term funds
- Savers demand convenience
- Solution: banks issue demand deposits and fund long-term loans

Let's stick with banks for now and unpack these points. The first issue savers and borrowers would run into is what we call maturity mismatch. On the one hand, borrowers typically want long-term funding, home buyers want mortgages for 15 or 30

years, corporations want funding for projects with long payback periods and on the other hand as savers, we want easy access to our funds and even value payment services. We most certainly do not want all of our money locked up for long periods of time. How do banks help to resolve this mismatch in needs? Well, they simply issue deposits that are redeemable on-demand and grant loans that are long-term.

2. Price risk

- Security prices/trading costs volatile
- Insufficient “safe assets”
- Solution: banks issue riskless deposits

Next up is the observation that stocks and bonds might not be the most stable store of value. Prices of stocks bounce around from day to day, the fees or bid-ask spread associated with trading might be volatile as well. When we invest in financial markets, these are usually the first issues that come to mind. Some savers might find price risk very off-putting and prefer just to hold cash instead. How do banks help here? Well, once again we can think of the deposit contract. It is both super convenient and also very safe. I put a dollar in the bank today and in a world with deposit insurance provided by the government, I will be able to withdraw at least one dollar tomorrow. There might be some fees associated with bank accounts but at least there is zero price risk.

3. Information Asymmetry

- Households cannot evaluate risk/returns

Final point and this is a big one, savers might have a tough time evaluating the risk and expected returns on investments. How much will the investment payoff? What is the likelihood of failure? How big is my loss in the event of default?

“Zano” the Baby Drone

- Europe’s most successful Kickstarter project at the time (>\$2m funding from >12,000 investors)
- Torquing group declared bankruptcy in November 2015

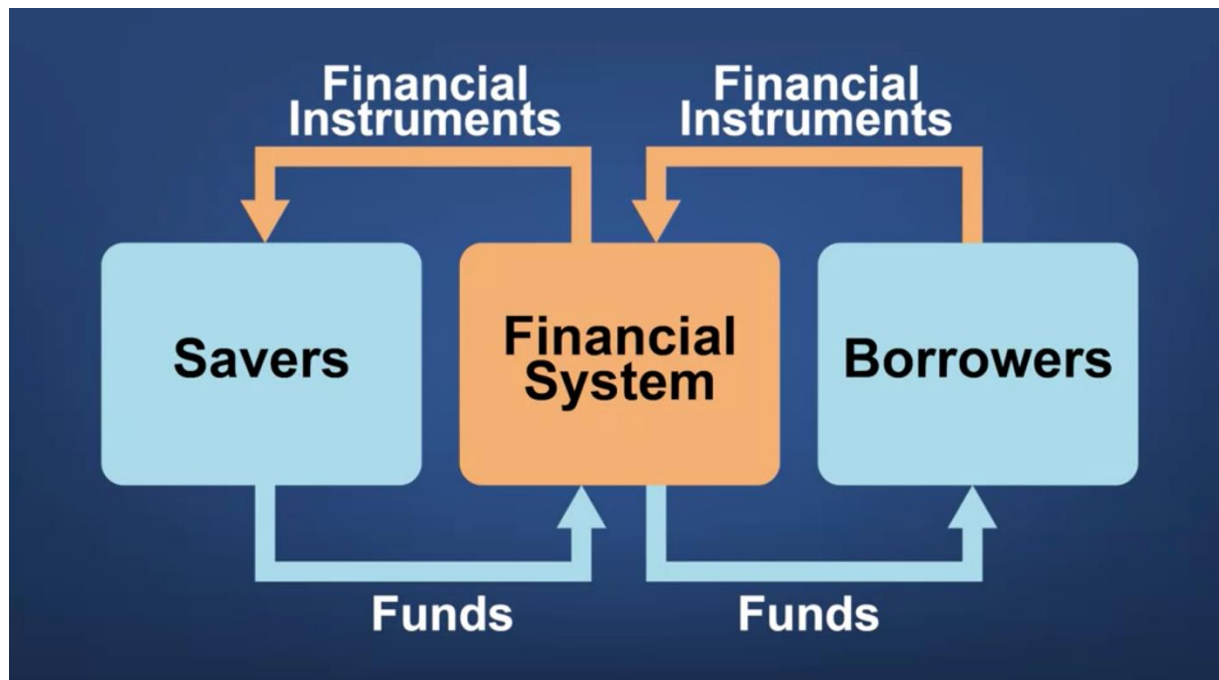
Here is an interesting case study. Zano, the baby drone, sounds like a really cool idea, a baby drone for selfies. Investors certainly agreed back in 2015 this award winning idea was Europe's most successful crowd-funded project. It had raised over two million dollars of funding for more than 12,000 investors on the Kick-starter platform. Unfortunately, it did not work out. Despite promises, only about 600 drones were

produced that could not fly and the parent company declared bankruptcy. Many of the investors ended up with absolutely nothing. What do we learn from this case? Well, it can be tough for individual investors to first select successful businesses and second minimize losses when things go bad.

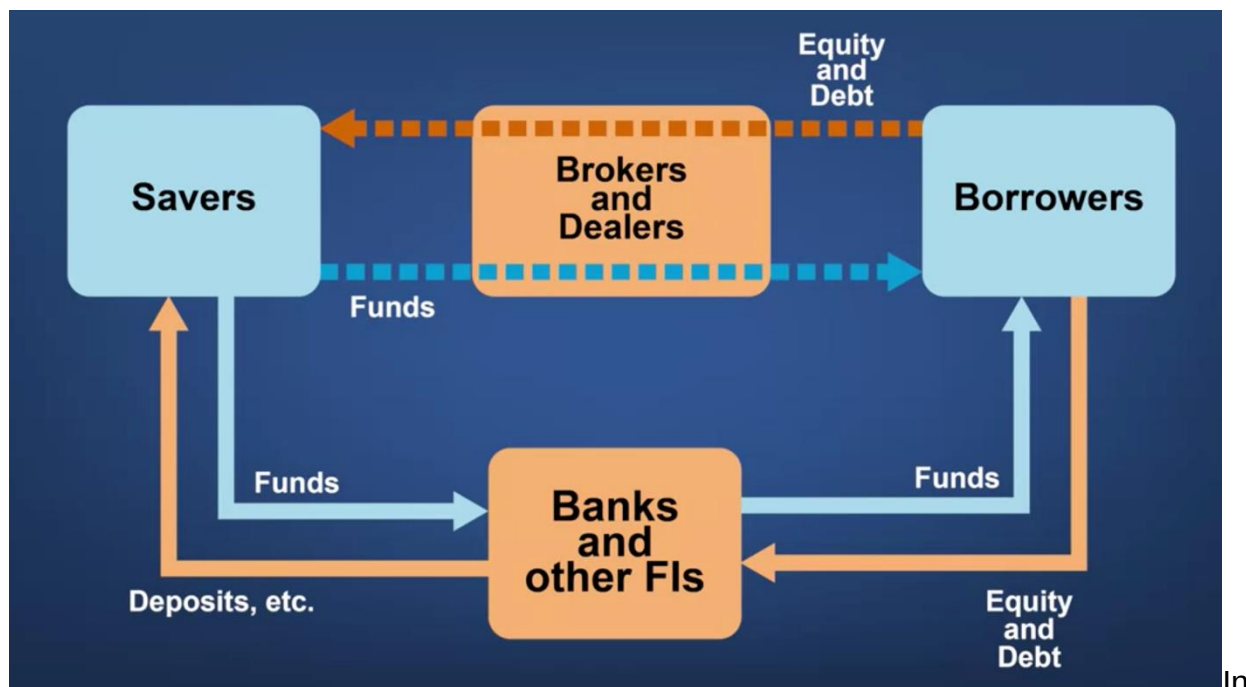
3. Information Asymmetry

- Households cannot evaluate risk/returns
- Market-based response insufficient
- Solution: banks screen/monitor borrowers

There are, of course, market-based solutions to these information problems. Equity research analysts forecast earnings, credit ratings agencies rate bonds, you can even buy bond insurance to guard against potential losses. However, if information risks are too large then some households might prefer not to invest. This creates a role for our bank, who can meet this information risks head on. Banks are experts at screening and evaluating borrowers, banks monitor borrowers after granting loans and banks stop lending to borrowers that are not performing well.



Let's summarize what we've just covered. The financial system moves funds from savers to borrowers and it does so through one of two channels, first, directly via financial markets.



In this case, a broker usually helps connect investors that can trade with each other. That way a saver can purchase securities issued by a corporation or the government. Second, funds are moved indirectly with the help of financial institutions. We have

argued that financial institutions can add value by solving maturity mismatch, price risk and finally information problems.

Lesson 1-2.3: Banks Versus Non-Banks

Learning Objectives

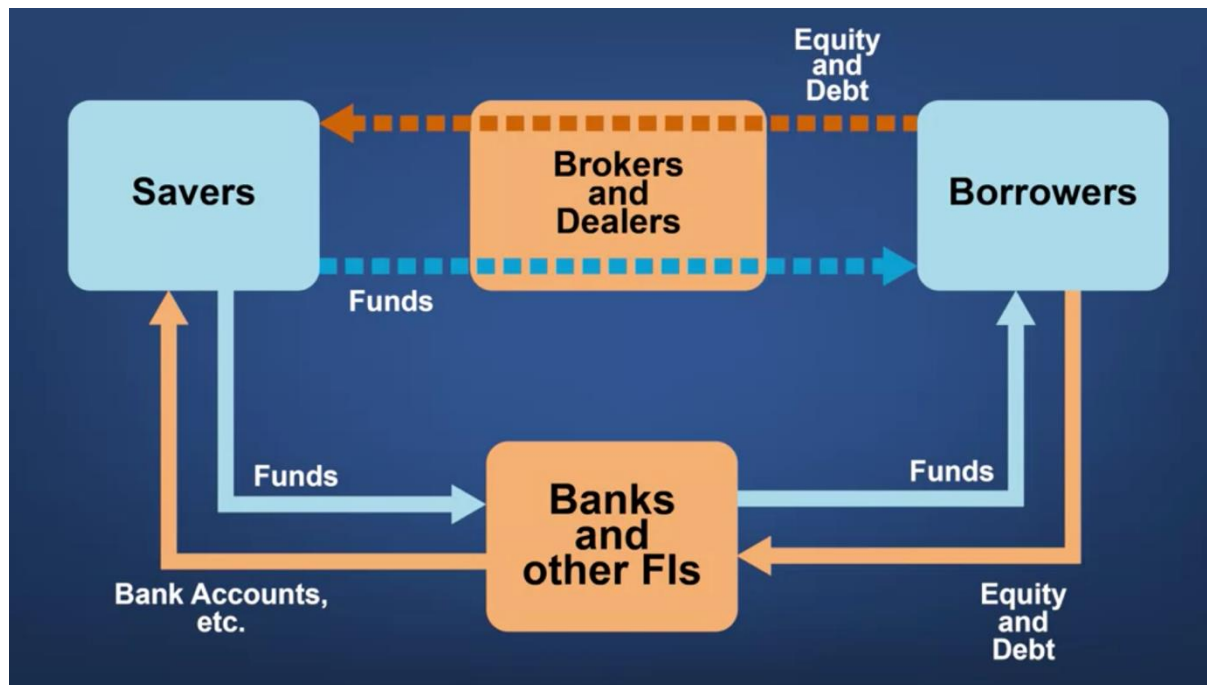
- Organization of financial industry

In this session, we will describe how the US financial industry is organized. We will introduce the most important bank and non-bank financial institutions and explain their roles in the economy.

Modern Financial Industry Structure

- Financial markets versus financial institutions

The modern financial industry consists of financial markets and financial institutions.



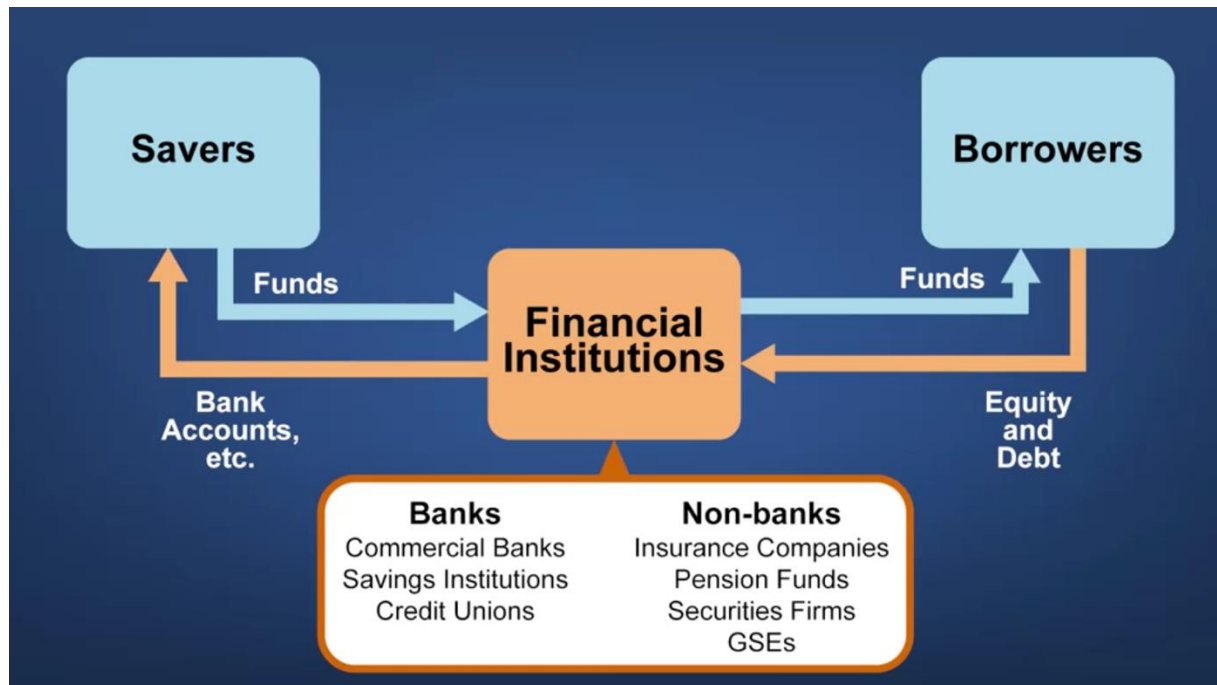
Some firms offer brokerage services that connect savers and borrowers directly in financial markets. Other financial firms, we have been calling them financial institutions, raise funds from savers and use the proceeds to invest in a variety of financial instruments: stocks, loans, bonds, and real estate, these financial institutions have balance sheets.

Modern Financial Industry Structure

- Financial markets versus financial institutions
- Key financial institutions

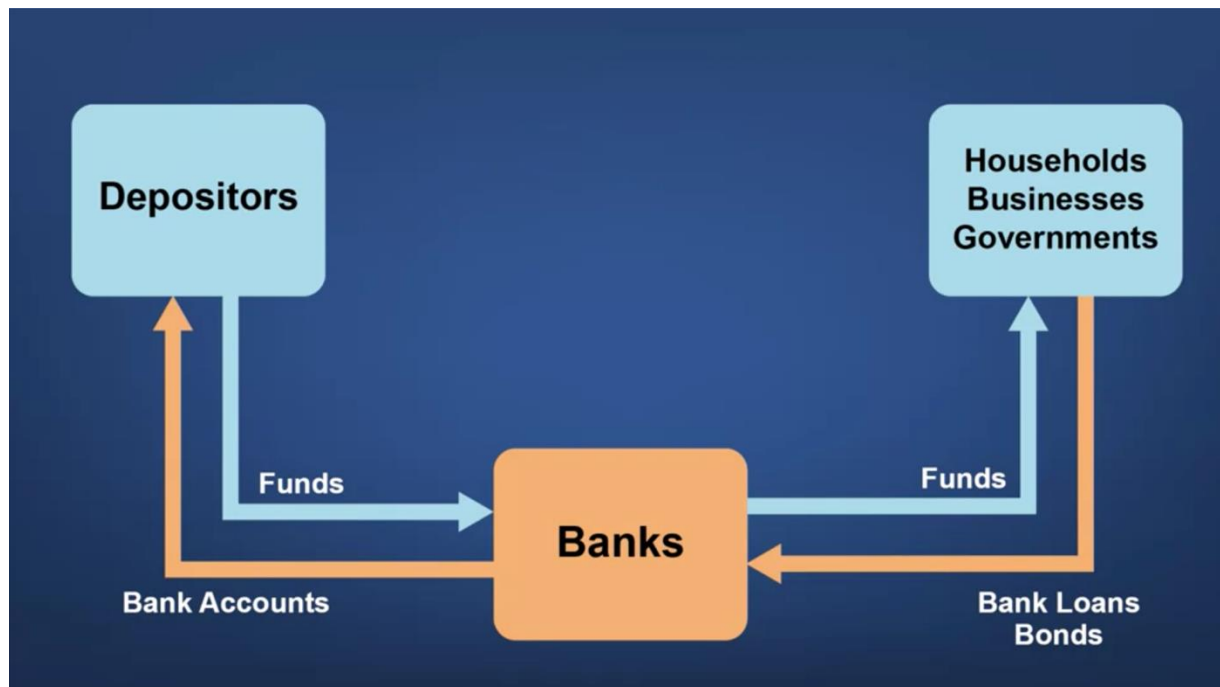
Our goal in this session is to identify and characterize the most important financial institutions. Who are they? What are their unique economic functions? How do they

raise funds? Where do they deploy these funds? Put differently. What do their balance sheets look like? And finally, which of these institutions matter most in terms of market share?

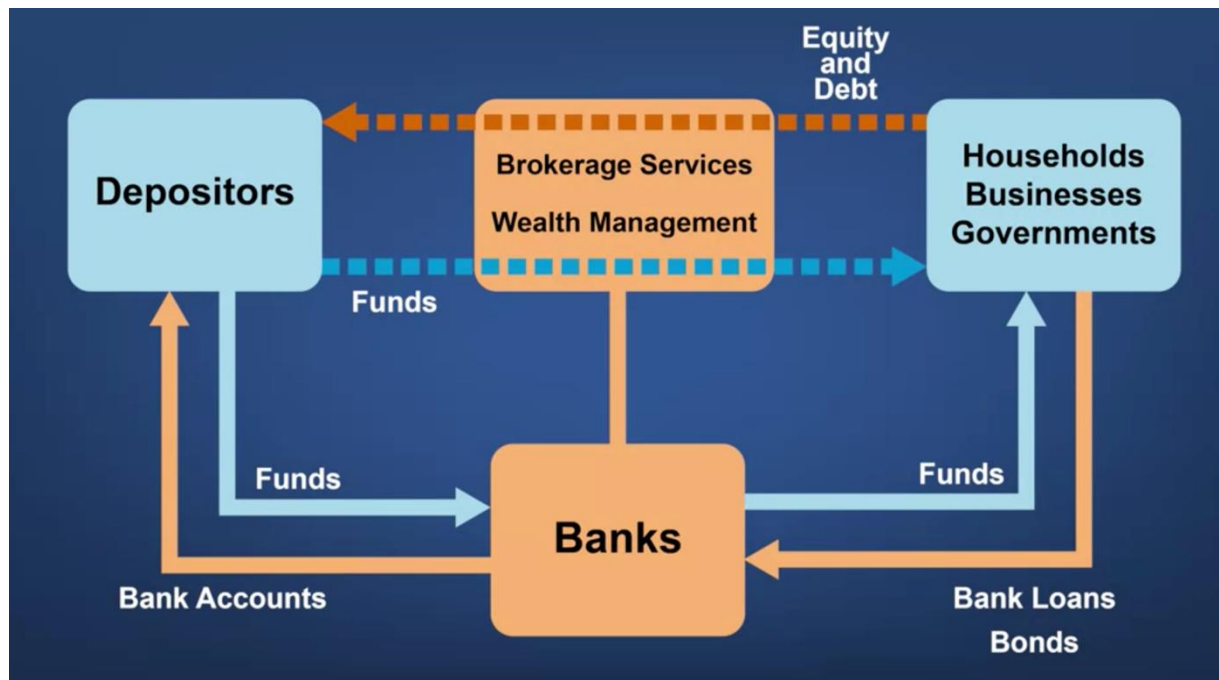


Let's open up the financial institutions box. We first make the distinction between banks and non-banks. Banks are the financial institutions that we interact with on a day-to-day basis. These include commercial banks, savings institutions, and credit unions. In different countries there are other types of banks. But they all do the same thing, they collect deposits from savers and grant loans to borrowers.

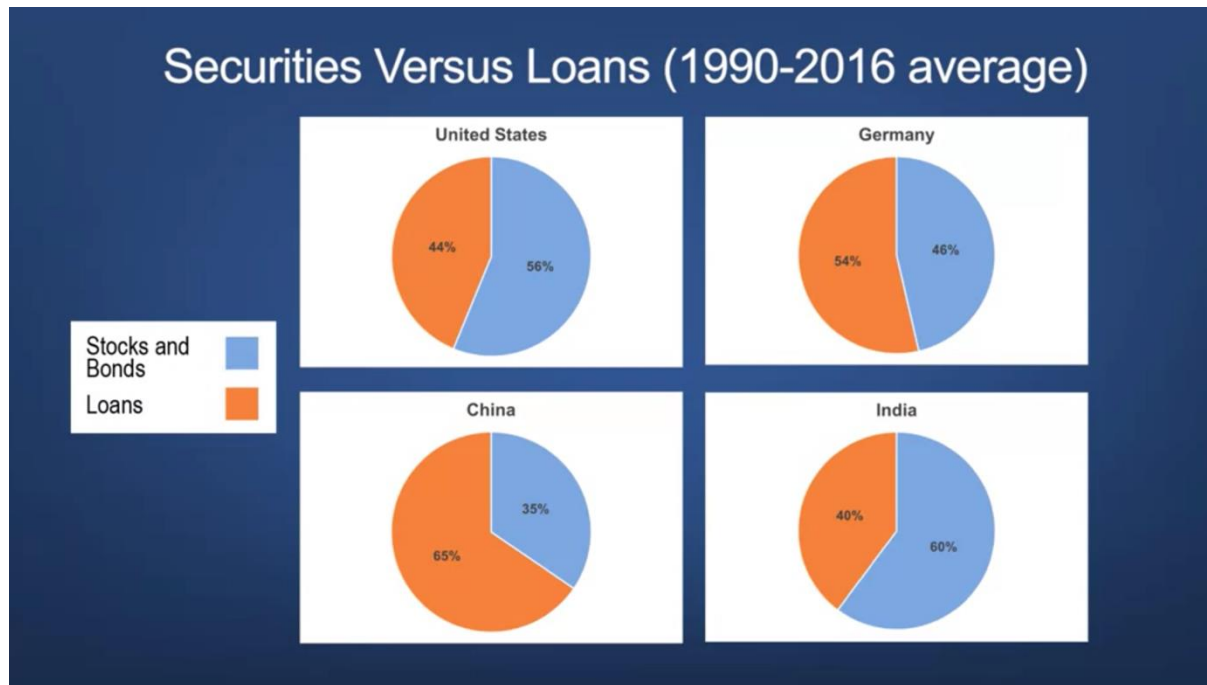
Banks are not the whole story here, we are going to see that non-banks play an important role and their presence has undoubtedly been increasing. What is a non-bank? Well, at a first pass, these are financial institutions that do not take deposits. The key types in the United States in terms of assets under management include insurance companies, pension funds, securities firms, finance companies, and government-sponsored enterprises. This is a very big umbrella and these institutions are very different. They have varied funding profiles, varied investment profiles, and the underlying economics of why these institutions exist also varies greatly.



Let's start with banks. We have commercial banks, savings and loans institutions and credit unions they all play a similar role in finance and their balance sheets reflect this. They raise funds primarily from retail customers. These are savers like you and me in the form of deposit accounts. These are our bank accounts. Some large banks also borrow in debt markets. You may have heard of the interbank market or the repo market. How did they use these funds? Well banks mostly invest in loans. In fact, banks are the primary originators of loans in the economy. Banks decide which households get a mortgage, the size of the loan, and the interest rate. What makes bank special? Why do we like banks? Well, banks are a combination of businesses. Depositors – that's us. we get bank accounts that are safe, convenient, and offer access to the payment system. Banks themselves pull our savings, in fact, large amounts of small savings and make loans to creditworthy borrowers. Finding those credit worthy borrowers and granting them loans is what we mean by origination.



Banks also trade in financial markets on behalf of their customers. This allows us to diversify our risks and manage our wealth into the future. Let's take a look at some recent data from the World Bank. We can calculate the size of domestic securities markets. This is stock market capitalization plus the total value of outstanding bonds and compare this with total loans outstanding. This comparison will give us a rough sense of the relative importance of banks, since banks grant the bulk of loans, as compared with securities markets all around the world.



Here is the data for the United States, Europe, and two Asian economies over the past quarter century. What are the key takeaways? First, in all of these economies, both securities and loan markets are important. This means that banks are highly relevant which is probably not surprising to you. Second, there are a lot of differences around the world. In the United States, finance is less dependent on banks whereas in China, the reverse is true. Moving onto our non-banks. First fact. Non-banks are not funded by government in short deposits. This is probably the loosest definition that I can offer.

Non-Banks

- No deposit liabilities
- Similar economic functions
- Vast array of types

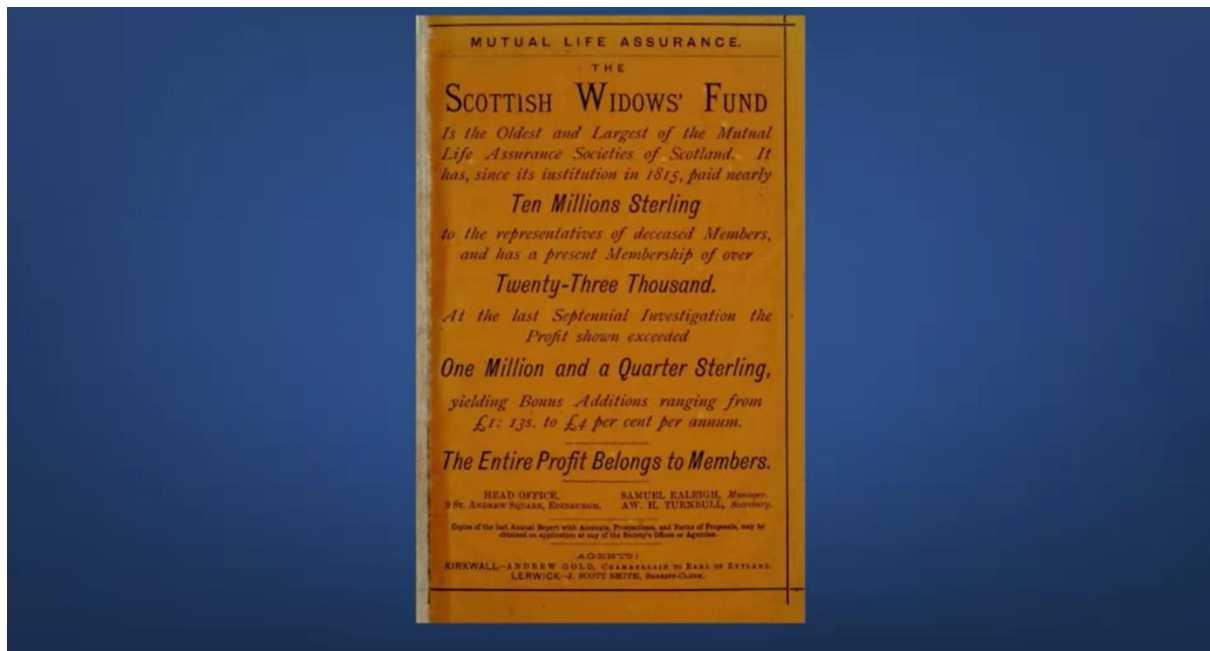
These non-banks perform many of the same economic functions as regular banks. They can provide a safe and convenient way to store wealth. They channel pools of small savings to borrower, they minimize information costs associated with investments, and some of them allow customers to reduce risk. There is a vast ecosystem of non-banks from peer-to-peer lenders to pawn brokers to private equity firms and of course we have the rise of Fintech in recent times.

[Lesson 1-2.4: Overview of Non-Banks](#)

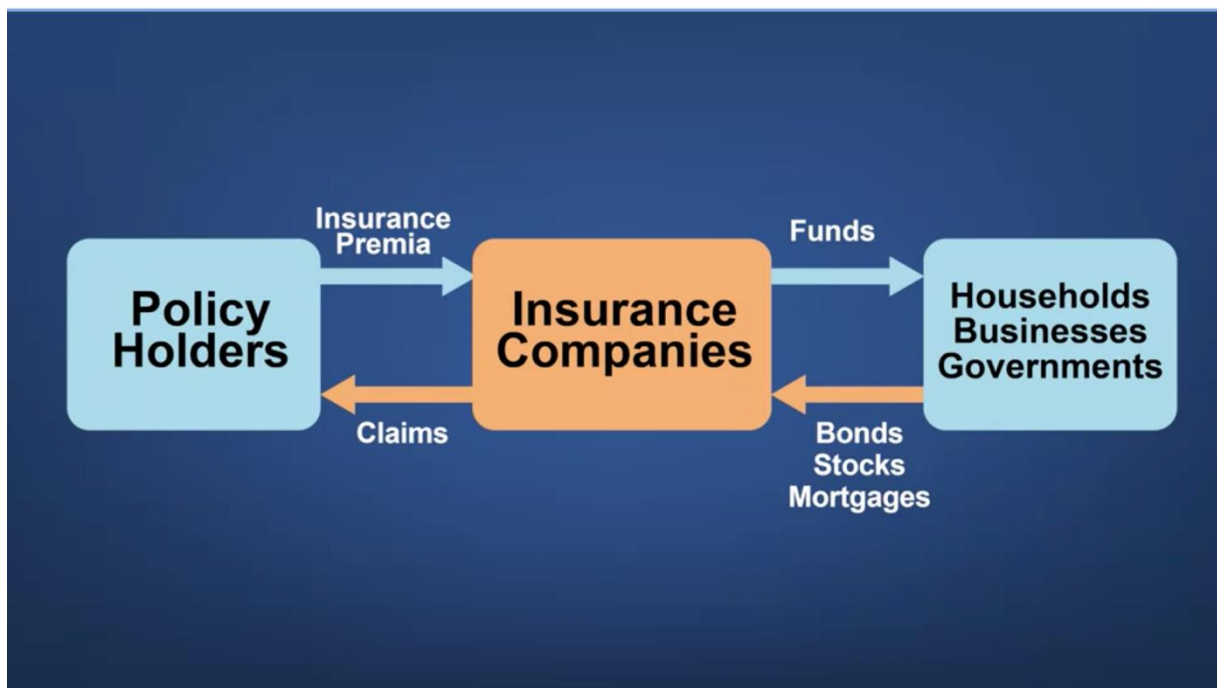
Insurance Companies

- Risk mitigation
- Life versus P&C insurance
- Balance sheets

We are going to focus our attention on the most important non-banks in the United States in terms of assets under management. First off insurance companies. The economic function of insurance is risk mitigation. From my perspective, it is a form of saving for a rainy day. I transfer the risk of some adverse event to an insurance company. I pay a premium. And then in the bad state of the world, I make a claim and the insurance company pays out. From the insurer standpoint, payouts become predictable as they write more and more and more policies. There are many insurance markets, but the two most common are life and property and casualty. With life insurance I'm protecting against the potential loss of earnings from disability or death.



My favorite example of life insurance is Scottish widows. This was a policy set up in the 19th century in Edinburgh, Scotland, to provide income to the families of men serving in the Napoleonic wars should they die in service. Property and casualty insurance protects households and businesses from losses due to natural events. You can think of floods and fires or due to accidents.



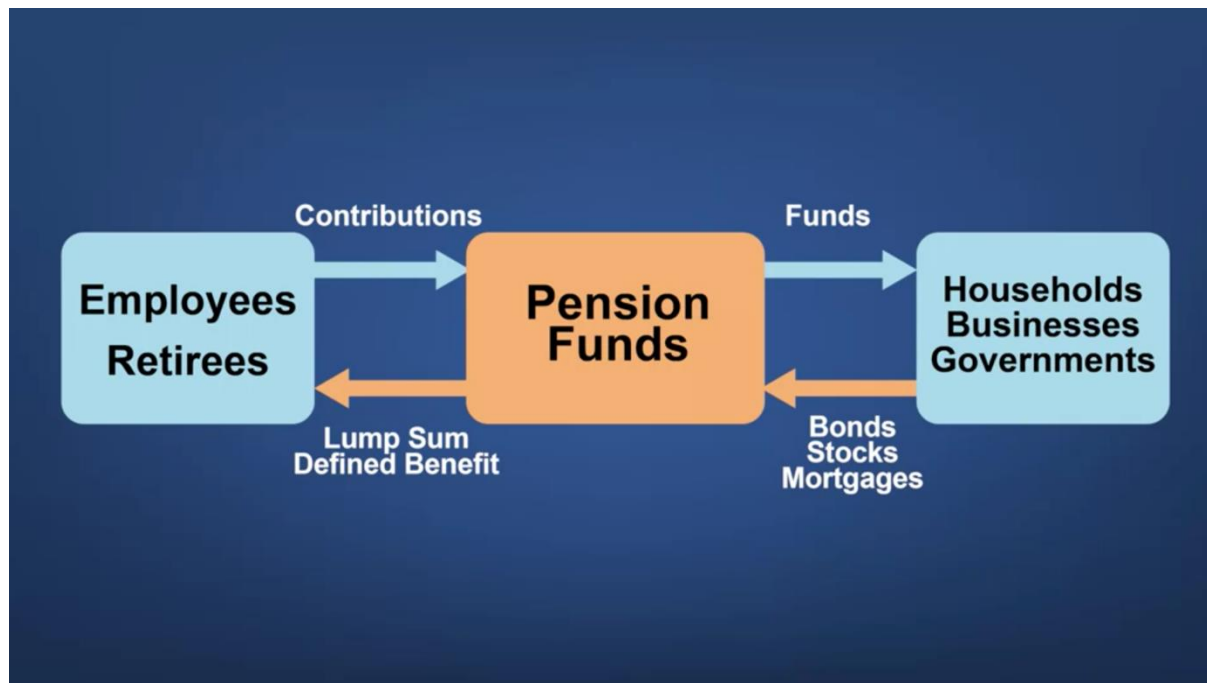
A good example is a car accident. What may be surprising is that these insurance premier are pulled and invested. My saving for a rainy day is channeled to borrowers.

More precisely, the source of funds for an insurance company consists of pulled premia. Insurance companies use these funds to make payouts on insurance claims. However, they also make long term investments in stocks and real estate and bonds. You might not have known that insurance companies are among the biggest investors in US corporate bonds.

Pension Funds

- Retirement savings

Next we have pension funds. Pension funds are used to store and grow our wealth into the future.

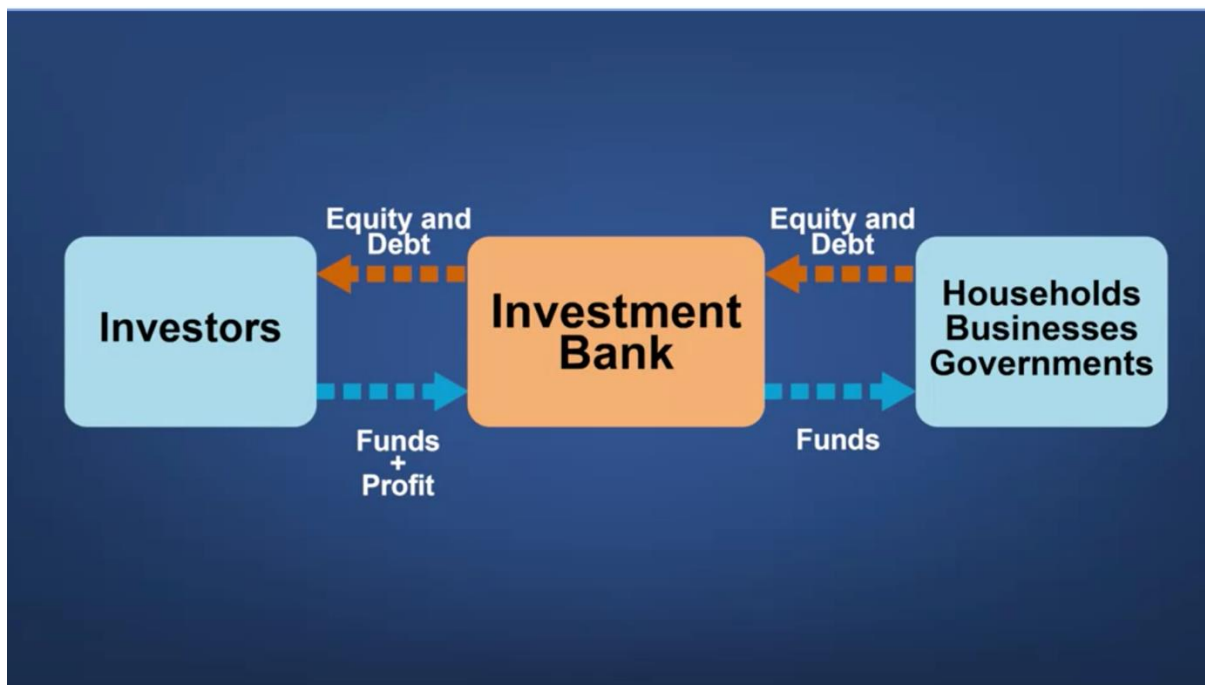


Households can save for retirement beginning at a young age and often in a way that is pre-committed. In my case, some 8% of my paycheck is set aside whether I like it or not. Retirement savings accounts usually offer some tax benefit, such as a deferral of income tax. Sometimes employers make contributions to these accounts. Once I reach retirement, I receive a payout from the fund usually in the form of a lump sum or a life annuity. Pension funds source their funds by pulling employee and employer contributions, and then the pension fund uses these funds two ways. It makes long term investments primarily in bonds, stocks and real estates. At the same time, it manages these investments to make payouts to retirees. Pension funds and life insurance companies are in a very similar line of business. Pension funds make payouts if you make it to retirement, whereas life insurance payout if you do not. Sorry if that sounds a bit morbid. Given these similarities these businesses tend to make the same investments. We also see the larger life insurance companies managing pension funds and vice versa.

Securities Firms

- Investment banks
- Investment funds

There are two main types of securities firms. First, we have investment banks. While commercial banks decide which households get mortgages and at what interest rates, investment banks decide which firms can issue securities and capital markets and on what terms.



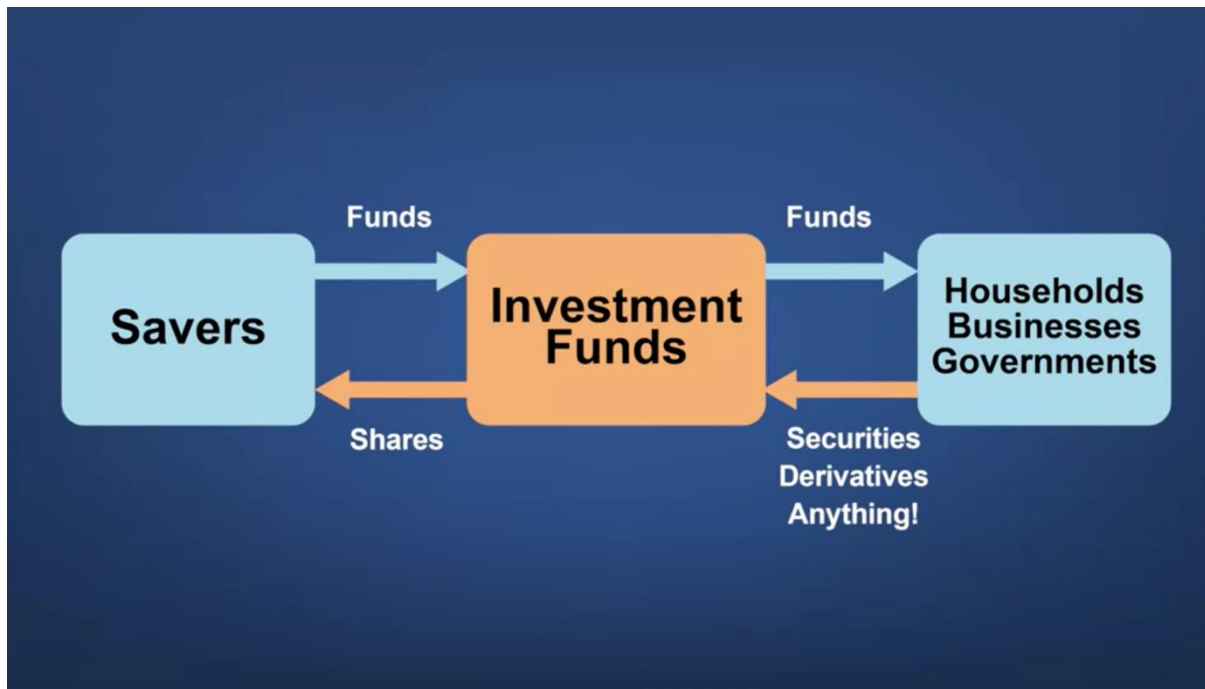
Underwriting is the bread and butter of investment banking. Let's say a corporation wants to issue some securities. The investment bank will commit to purchasing the

entire issue at some price. It will then place the issue with outside investors shortly after, hopefully at a higher price and yielding some profit.

Securities Firms

- Investment banks
- Investment funds

Taking a step back, investment banks have two economic functions. First, they are in the business of collecting and processing information about potential borrowers. They act as gatekeepers for capital markets. Second, they connect savers and borrowers through the underwriting process.



Investment funds are very different. They pull savings and make investments in large diversified portfolios of assets. This allows investors to store and grow their wealth into the future. There is a wide variety of mutual funds and exchange traded funds that serve this purpose. Retail investors can access these funds with minimum investment requirements and low transaction fees. They have a variety of investment profiles. Money market funds only invest in super safe and short term bonds. Equity mutual funds are riskier. The polling part is very important. Vanguard, which is one of the largest mutual fund families, will offer a mutual fund that tracks the S&P 500 with a minimum investment of \$3,000. It would be very challenging to replicate this investment without their help. Hedge funds are an infamous class of investment fund. They cater to wealthy and sophisticated investors only. A hedge fund can either have 99 investors, each with a net worth in excess of \$1 million dollars or 499 Investors worth more than \$5 million dollars apiece. Avoiding uninformed retail investors, that's you or me allows hedge funds to sidestep certain regulations. This means hedge funds can pursue high risk, high reward investment strategies across securities and derivatives markets and face very little scrutiny.

Finance Companies

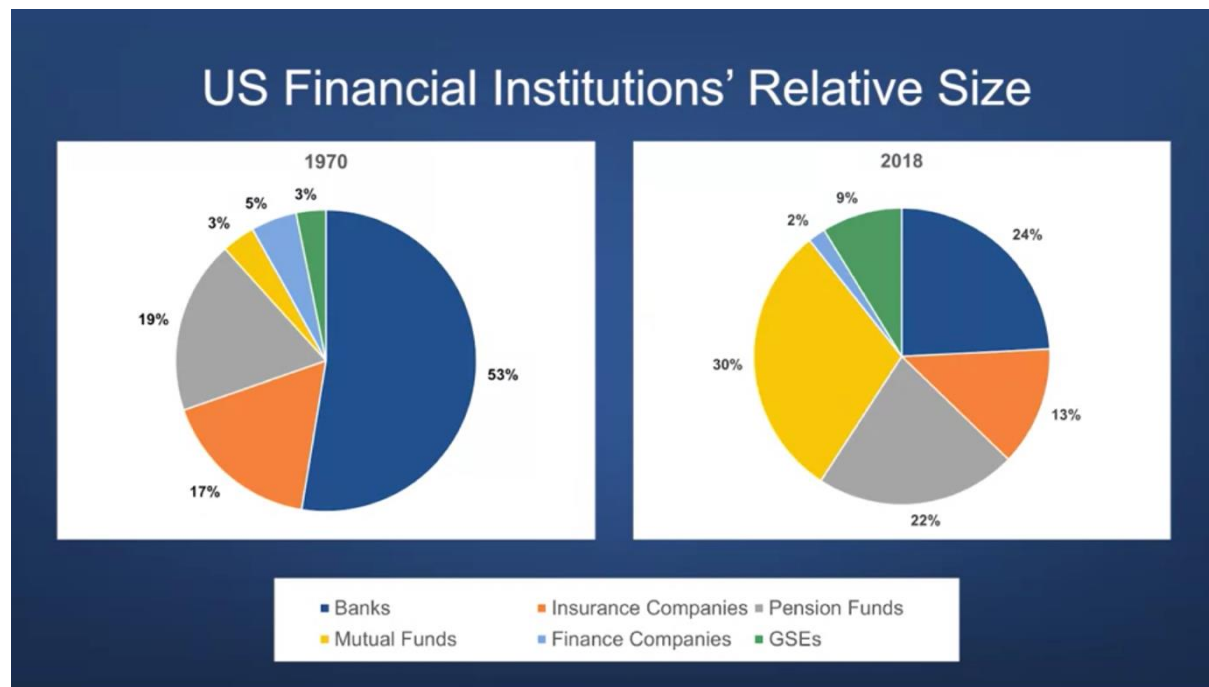
- Specialized lenders
- Minimize information costs
- Banks without deposits

Next, we have finance companies. These financial institutions look a lot like banks. Only they cater to a specific loan market. Some focus on consumer credit, which is credit for household purchases. This could be a mortgage to buy a house, an installment loan for an appliance like a fridge or a washing machine, or an auto loan to buy a car. For example, the Ford Motor Credit Company finances car sales. Other finance companies specialize in business lending. This could be anything from financing and aircraft purchase to providing a lease for farm equipment. Finance companies act a lot like banks, but with a specific industry focus. Granting loans and leases is their primary activity just like banks and they minimize information costs by screening and monitoring borrowers. However, they're not banks, and their fundraising activities reflect this. Finance companies are not funded with deposits. They are instead funded with bonds and bank loans.

Government-Sponsored Enterprises (GSEs)

- Fannie Mae, Freddie Mac, and Ginnie Mae
- Subsidized mortgages
- Balance sheets

Our final major class of non-bank is the government sponsored enterprise, or GSE. These are key players in the United States mortgage market. Who am I talking about? Well, there are three financial institutions here. The Federal National Mortgage Association, also known as Fannie Mae. Fannie was established in the 1930s, the Federal Home Loan Mortgage Corporation. This guy is Freddie Mac and it arrived on the scene in 1968. Finally, we have Ginnie Mae or the Government National Mortgage Corporation, which also arrived in 1968. All of these financial institutions were established by the US government. Ginnie is wholly owned by the government. Fannie and Freddie are private corporations, although since they were established by the government to serve the public good, they have been dubbed GSEs that is government sponsored enterprises. The GSEs support home ownership in the United States, especially among lower income and underserved communities. This includes loans for low cost housing or loans to army veterans. The GSEs achieved this by subsidizing mortgage interest rates. This happens either through direct lending at below market rates or by guaranteeing repayment on certain real estate financial instruments issued by other financial institutions. These guarantees act a lot like insurance, which makes these subsidized loans less risky. Putting this together, the GSEs raise funds by issuing bonds or selling loan guarantees to other financial institutions. And they use these funds two ways. First and primarily, they grant mortgages. Second, they make insurance payouts related to their loan guarantees.



Okay, we have covered the major financial institutions in the United States. Our final task is to examine the relative importance of these institutions. To do this, we will look at data from the US flow of funds. These data are collected and processed by the Federal Reserve. They contain estimates of the asset holdings of banks and the major groups of non-banks. We are going to examine two different time periods. 1970 or as I like to call it, ancient history and 2018, which is more or less today. What do we see? Well, two things stand out. First, the declining share of banks among these financial institutions. In 1970 banks funded more than half of all assets. By 2018 this had declined to less than 25%. Insurance companies also experienced a decline. Who stepped in? Clearly mutual funds. Their share of assets has increased from less than 5% to more than 30%. This reflects the increased access to low cost index investing through products offered by Vanguard and other investment funds.

Summary

- Financial institutions connect savers and borrowers
- Recent trends towards non-banks

Let's sum up, financial institutions channel funds from savers to borrowers. There are many bank and non-bank financial institutions involved in this process. We have characterized these institutions in terms of their key economic functions and balance sheets. Data indicate that banks play an important role in financing around the world. In the U.S., however, non-banks especially mutual funds seem to have become prominent in the recent years. At this point, we have not discussed any potential implications. Good or bad of this reallocation. That is for another time.

Lesson 1-3: Finance and the Real Economy

[Lesson 1-3.1: Financial Development and Economic Activity](#)

Learning Objectives

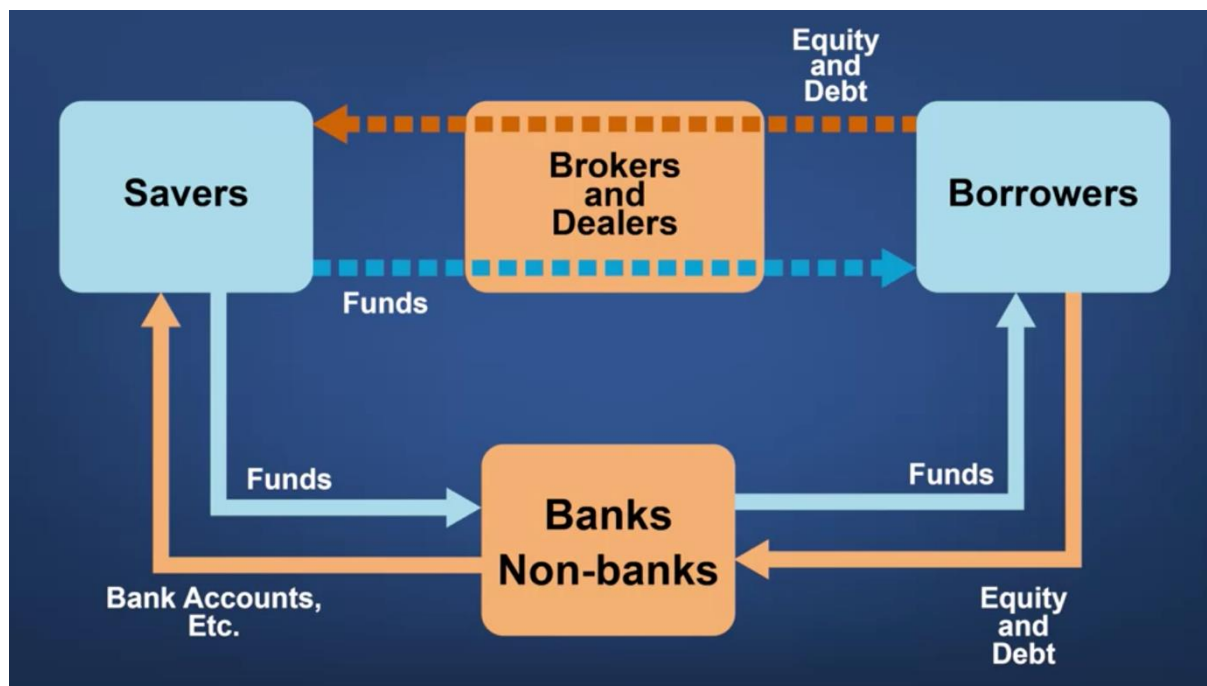
- Financial development and economic growth
- Financial crises

Money and finance are central to a well-functioning economy. In this session, we will provide evidence that greater financial development leads to better economic outcomes for society. We will then argue that it's not always smooth sailing. Sometimes there can be disruptions within the financial system. This can occur when investors lose faith in financial institutions or when financial innovations are too rapid and go unchecked by markets or regulators. In the second part of this session, we will introduce the idea of a financial crisis and discuss how these events can be costly for society.

Modern Money and Finance

- Money and finance

A well-functioning economy depends on money and the payment system.

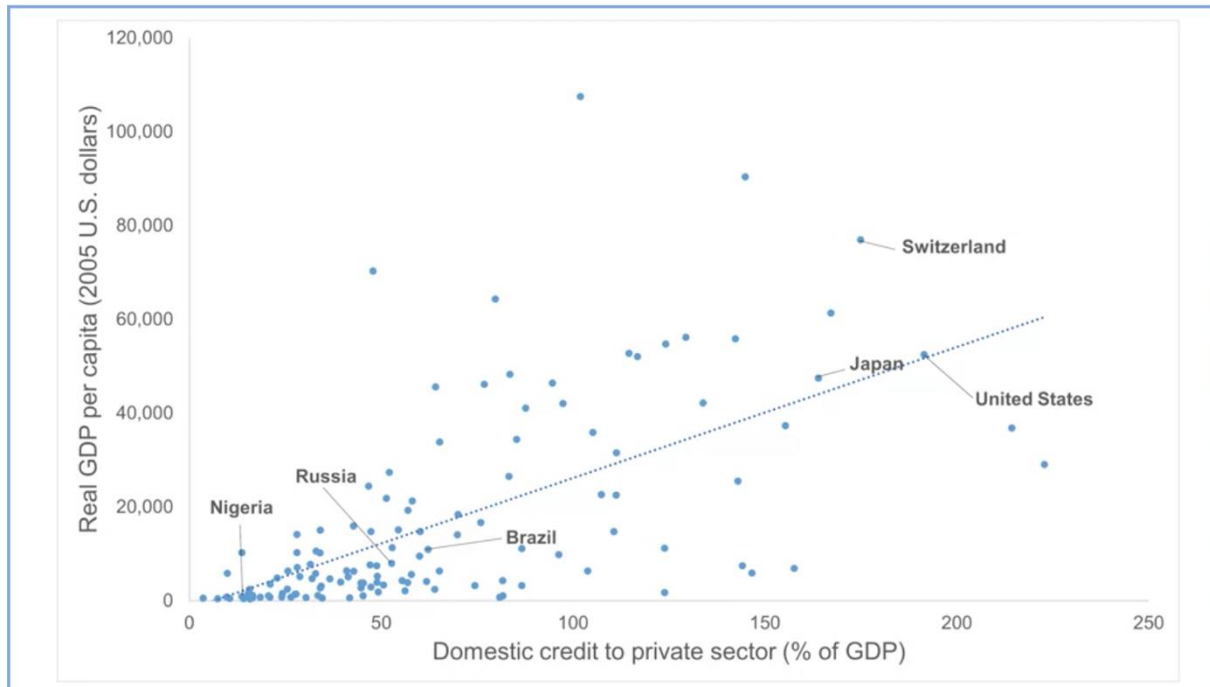


The financial system, which involves financial market and a broad array of bank and non-bank financial institutions, channels funds from savers to borrowers. This promotes investment. Our belief is that greater financial development means more accurate market price and more efficient allocation of resources and better risk sharing.

Modern Money and Finance

- Money and finance
- Economic efficiency

You put all of that together and we should see faster economic growth and better outcomes for society. We can test this idea using recent data from the World Bank. We can look across the globe and check whether countries with larger financial systems have better economic outcomes. To measure financial development, we can take the total amount of credit extended to the private sector. We have to scale credit by GDP so that our observations across countries are comparable. We then focus on GDP per capita, as a measure of economic development.



The scatter plot seems to confirm our intuition. We can see that countries with larger financial systems tend to have higher incomes per citizen. That's great news. And it suggests that we want to have more and more financial development, more and more banks, more financial markets and possibly fewer restrictions on these institutions, right? If you're a policy maker, then you should prioritize the financial sector above many other things. But surely there is a limiting principle here.



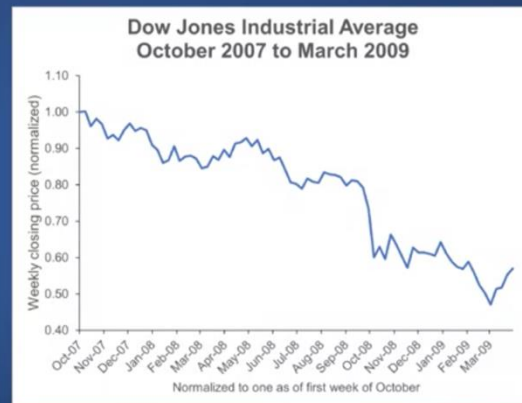
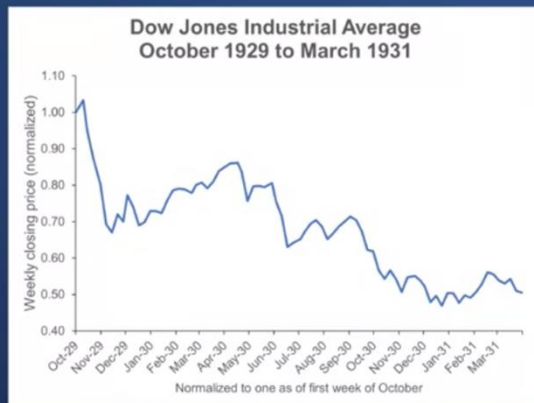
Well, it's hard to say financial without saying financial crisis. Throughout history, we

have seen episodes like this. A depression era bank in the 1930s, with customers queuing up hoping to withdraw their life savings.



Northern Rock, a Bank in England, where I grew up, experienced a similar run on its deposits in the fall of 2007. This was an early warning sign that a financial crisis was on its way. Whether it's the Great Depression or Great Recession, we seem to hear the same narrative. In a first step, excesses within the financial system cause a panic. Savings are withdrawn from financial institutions, and then these institutions are forced to liquidate investments.

Stock Market Crashes



Eventually financial markets such as the stock market crash. Then in a second step, this financial crisis causes a terrible burden on the economy. Households lose their savings, and they can no longer access credit. Businesses fail. All of this translates into a deep and long lasting recession. Banks were central to the original version of this story.

History of Banking Panics

“Since 1793 panics have occurred [in the United States] in the following years: 1797, 1811, 1813, 1816, 1819, 1825, 1837, 1847, 1857, 1866, 1873, 1884, 1890, and 1893. The true name for a system with this record is the panic system.”

Theodore Gilman, Federal Clearing Houses, 1899
Reprinted in Gary B. Gorton, Misunderstanding Financial Crises, 2012

Since 1793 panics have occurred in the United States. In the following years, 1797, 1811, 1813, 1816 and so on and so on. 1873, 1894, 1890, 1893. The true name for a system with this record is the panic system.

History of Banking Panics

“The Panic of 1907 was the first worldwide financial crisis of the twentieth century.”

Moen and Tallman, Federal Reserve History, 2015

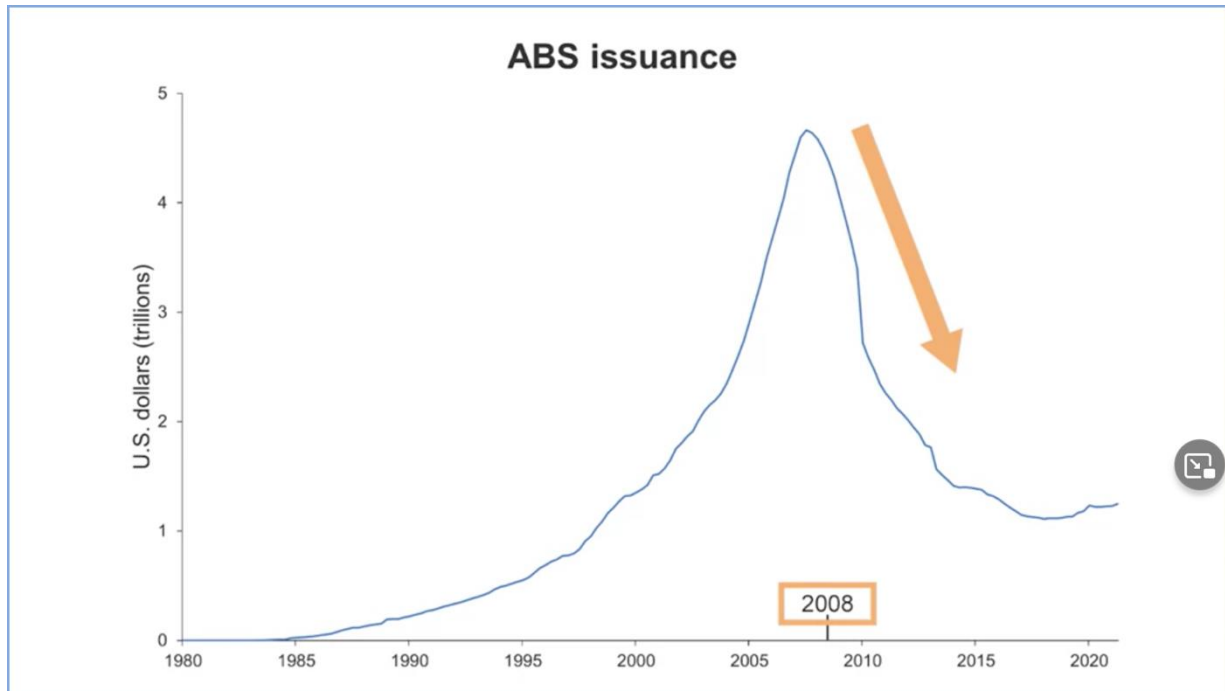
The panic of 1907 was the first worldwide financial crisis of the 20th century. These banking panics peaked during the Great Depression of the 1930s. Between 1930 and 1933, about 2000 banks failed per year. This was probably the worst global financial crisis in modern history. It led to the creation of our current system of government deposit insurance provided by the Federal Deposit Insurance Corporation, or FDIC. 100 years later, financial crisis still involve panicky investors.

Panics in the 21st Century

- Securitization
- Runs at money market funds

Let's start with Securitization. It is one of the great financial innovations of the 20th

century. It allows financial engineers to take pools of risky loans and create new asset backed securities or ABS. Some of these ABS are rated Triple A, so we have transformed risk into safety.



However, the process itself is complex and opaque, so it can be hard for many investors to understand. Beginning in 2007, some of the riskiest mortgage backed securities. You may have heard of subprime mortgages before they started to default. Investors didn't think that was possible, and they panicked. The panic spread to all securitization markets, including student loans and credit card securitizations that had absolutely nothing to do with mortgages. There was a near total collapse in credit, money market funds. They have been central to financial panics in the 21st century. They are a type of mutual fund that makes super safe investments in short term bond markets. Large institutional investors park cash at these funds to earn some returns, small returns, but better than nothing. Here's the fun part. These funds structure their liabilities to resemble bank deposits. They appear riskless and can be redeemed on demand. But savings in money market funds are not insured by the government and may be subject to losses. In 2008, when losses derive at one very famous fund, you probably don't know it, but that fund was called the Reserve Primary Fund. This was totally unexpected. Investors panicked and the entire money fund industry experienced mass withdrawals and was brought to the brink of collapse. Problems can stem from the money and payment system, too. Hyperinflation in Weimar, Germany in the 1920s, Hungary in 46, Zimbabwe in 2008, Venezuela in 2018. These are extreme examples.

Too Much Money?

- Money growth and inflation
- Economic collapse



In each case, governments issued too much paper money. The paper money was eventually deemed worthless. The outcomes for the economy and society as a whole were truly dreadful.

Too Much (Digital) Money?

- Digital currencies and financial stability
- “Stablecoins” and Tether

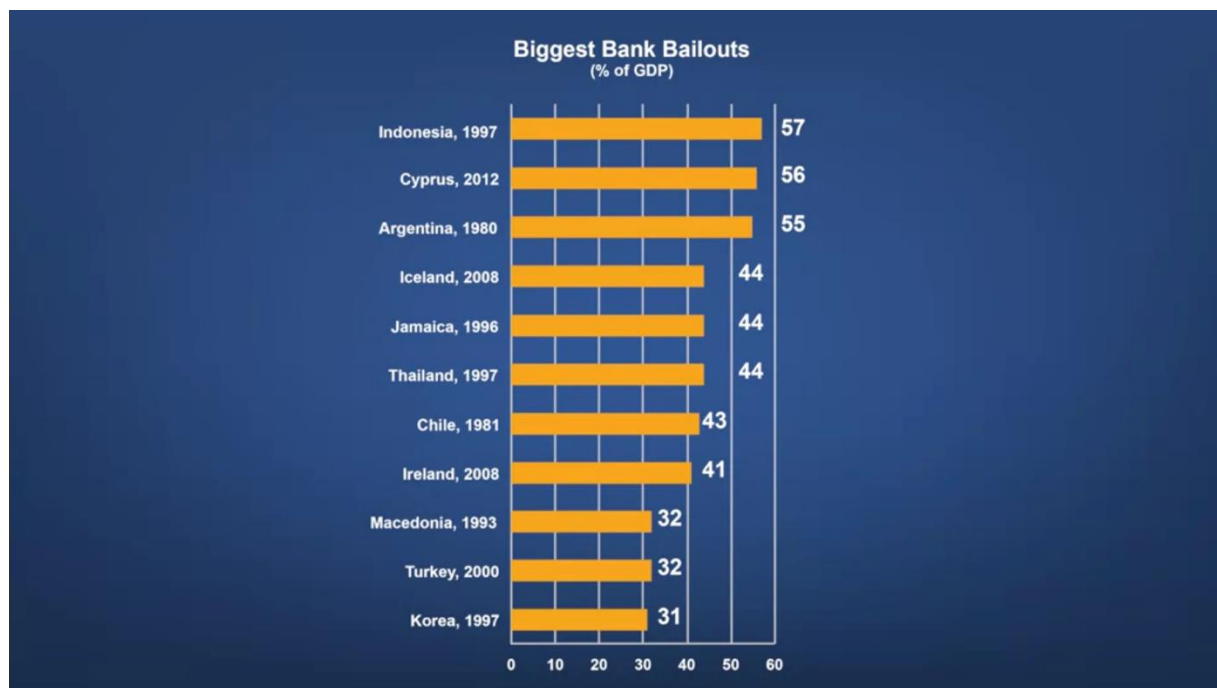
Digital currencies break government control over the supply of paper money. However, since the emergence of Bitcoin in 2008, they have looked more like a speculative bubble than a reliable means of payment. One point of view is that so long as price risks are dispersed among retail investors, then it probably doesn't matter if the value of digital currencies rise or fall. It is a problem for the investors, but not a threat to the

financial system and the broader economy. Stable coins are not the same as Bitcoin. Stable coins are a special type of digital currency that peg their value to a regular currency. Tether is the most popular stable coin. As of 2021, Tether has issued more than \$60 billion dollars' worth of digital tokens, and those digital tokens are redeemable for a dollar apiece. These stable coins can be used to pay for things, and they have value because they can be exchanged for dollars. You might not know this, but Tether actually invests the funds it raises by selling these digital tokens. Where? Well, from what we can gather it invest the dollars it collects in securities such as commercial paper. As of 2021, Tether had about \$30 billion dollars' worth of commercial paper investments, making it essentially one of the biggest money market funds in the world. What's going to happen if investors lose faith in Tether and redeem their digital tokens on mass? Well, if some investors panic and worry they might not be able to get \$1 per token, then they might be willing to accept 90 cents or something less. This could lead to a run on Tether, and the value of the digital currency could fall immediately. A panic of this sort could disrupt the entire stable coin market. Worse yet, and this is why it matters, this disruption could spill over to the commercial paper market. This could affect how governments and businesses finance their working capital. It sounds like some digital currencies might be more problematic than others.

Government “Bailouts”

When the financial system fails, time and time again, we have seen the government step in and provide support. This action is usually called a government bailout. In the United States, the various government agencies spent more than 600 billion dollars of taxpayer money during and after the 2008 financial crisis. That is a large sum, but the U.S. is also a very big economy. It can afford to spend that much money provided the

intervention succeeds, of course. Thankfully, it more or less worked out back in 2008. Other governments cannot support bailouts of that magnitude.



This chart shows the largest bank bailouts in recent history. We scale the size of the bailout by GDP to give us a sense of how costly it was relative to national income. In 1997, Indonesia spent more than half of its national income bailing out banks. Bailouts of that magnitude can lead to a sudden collapse of government finances. All right, where does this leave us? There is a tradeoff. We want to promote an efficient, competitive financial sector. Innovations in money and finance move our economy forward. But we need to balance this against financial sectors stability and the risk of crashes.

Taking Stock

- Efficiency versus resilience
- Regulation

To guard against crashes, we need to set limits. First, which institutions should be regulated. Should regulators leave private equity and FinTech firms alone? This is an open question. Second, what should regulated financial institutions be allowed to do? Should Tether have the same disclosure requirements as banks? Should banks be allowed to have hedge funds subsidiaries? Should we allow those hedge funds subsidiaries to invest in Bitcoin? Finally, what sorts of government interventions are appropriate during financial crises? Our intuition is that it's sometimes appropriate to bail out failing institutions. But how should limits be set? These are important public policy questions.